

Theory of active self-organization of dense nematic structures in the actin cytoskeleton

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eLife assessment

In this study, the authors offer a theoretical explanation for the emergence of nematic bundles in the actin cortex, carrying implications for the assembly of actomyosin stress fibers. As such, the study is a **valuable** contribution to the field actomyosin organization in the actin cortex. While the theoretical work is **solid**, experimental evidence in support of the model assumptions remains **incomplete**. The presentation could be improved to enhance accessibility for readers without a strong background in hydrodynamic and nematic theories.

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Abstract

The actin cytoskeleton is remarkably adaptable and multifunctional. It often organizes into nematic bundles such as contractile rings or stress fibers. However, how a uniform and isotropic actin gel self-organizes into dense nematic bundles is not fully understood. Here, using an active gel model accounting for nematic order and density variations, we identify an active patterning mechanism leading to localized dense nematic structures. Linear stability analysis and nonlinear finite element simulations establish the conditions for nematic bundle self-assembly and how active gel parameters control the architecture, orientation, connectivity and dynamics of self-organized patterns. Finally, we substantiate with discrete network simulations the main requirements for nematic bundle formation according to our theory, namely increased active tension perpendicular to the nematic direction and generalized active forces conjugate to nematic order. Our work portrays actin gels as reconfigurable active materials with a spontaneous tendency to develop patterns of dense nematic bundles.

Introduction

Actin networks are remarkably dynamic and versatile and organize in a variety of architectures to accomplish crucial cellular functions [1]. For instance, isotropic thin actin gels form the cell cortex, which largely determines cell shape [2] and motility in confined non-adherent environments [3, 4]. Polar structures at the edge of adherent cells, either forming filaments as in filopodia or sheets as in lamellipodia [5], enable cells to probe their environment and crawl on substrates. Nematic actin bundles conform a variety of contractile structures [6], including sub-cellular rings during cytokinesis [7] and cortical repair [8], supra-cellular rings during wound healing [9] or development [10], bundle networks during cellularization [11], or stress fibers in adherent cells [12–14]. Nematic bundles consist of highly aligned and densely packed actin filaments of mixed polarity connected by a diversity of crosslinkers. In vivo and in vitro observations show the key role of actin nucleators and regulators, of myosin activity and of crosslinkers in the assembly and maintenance of actin bundles [1, 2, 5, 6, 13, 15–21].

Various studies have emphasized the morphological, dynamical, molecular and functional specificities of different types of actin bundles such as dorsal, transverse, and ventral stress fibers or contractile rings [13, 14, 22–24]. Here, we ask the question of whether, despite this diversity, the ubiquity of actin bundles in different contexts can be explained by the intrinsic ability of the active actomyosin gel to self-organize into patterns of dense nematic structures. Suggestive of such active self-organization, stress fibers often form dynamic highly organized patterns, e.g. involving families of fibers along orthogonal directions [12, 14, 19, 22, 25–27]. Other kinds of actin bundles also develop patterns of remarkable regularity. For instance, parallel arrangements of 2 μm -spaced actin bundles serve as templates for extracellular matrix deposition during butterfly wing morphogenesis and determine their iridescence [28]. Similarly, the morphogenesis of the striated tracheal cuticle in *Drosophila* is pre-patterned by a parallel arrangement of actin bundles spaced by $\sim 1\mu\text{m}$, spanning from sub-cellular to supra-cellular and organ scales [21, 29]. Muscle-like actin bundles form regular parallel patterns spanning organs as in *C. elegans* [19] or the entire organism of hydra [30]. Furthermore, dense nematic bundles have been shown to assemble de novo from the sparse isotropic cortex in a process controlled by myosin activity [20], and to form a mechanically integrated network with the cortex [31].

Theoretical models of stress fibers often assume from the outset the existence of dense actin bundles, possibly embedded in an isotropic network [32], although previous work has studied the emergence of nematic rings as a result of localized activity gradients [33]. In the field of active nematics, many theoretical studies have examined the well-known hydrodynamical bend/splay instability arising in uniform active systems with pre-existing long-range orientational order [34–36]. However, the question of how patterns of orientational order arise from an isotropic and uniform system as a result of activity has received very little attention, and has been based so far on models that do not capture fundamental physical features of actomyosin gels as discussed below [37, 38]. Here, to understand the mechanisms underlying the self-assembly of dense actin bundles from a low-density isotropic cortex, we develop a theory for the self-organization of dense nematic structures in the actomyosin cytoskeleton based on a nematic active gel theory accounting for density variations and compressibility. In our theory, nematic patterning is driven by activity rather than by a more conventional crowding mechanism. Linear stability analysis and fully nonlinear simulations show that, when coupled to nematodynamics, the well-known patterning mechanism based on self-reinforcing flows [29, 39–43] leads to a rich diversity of patterns combining density and nematic order. The geometry and dynamics of

the emergent patterns are very similar to those observed in diverse cellular contexts. Finally, we test key assumptions of our phenomenological theory leading to such self-organization using discrete network simulations.

Theoretical Model

The actomyosin cytoskeleton can be understood as a compressible, active and viscous fluid gel with orientational order undergoing turnover [44, 45]. Symmetry-breaking and pattern formation in actomyosin gels is often mediated by the emergence of an advective instability leading to compressible self-reinforcing flows. According to this mechanism, fluctuations in the density of cytoskeletal active units generate gradients in active stress, driving flows, which in turn advect the active units reinforcing the initial fluctuation [29, 39–41]. This kind of advective instability has been invoked to explain cell polarization and amoeboid motility [4, 41, 46], the formation of the cytokinetic ring [42], the formation of periodic dense actin structures during tracheal morphogenesis in *Drosophila* [29], or the emergence of self-sustained dynamical states in actomyosin gels extracted from cells [47, 48], and has been recently reproduced to some degree in confined reconstituted gels from purified proteins [49]. In all of these examples, the actomyosin gel develops sustained compressible flows converging towards regions of high density. Furthermore, the compressive strain rate induced by these active flows has been shown to drive nematic order [7, 33]. Finally, observations on adherent cells show that active contractility is required for actin bundle formation [19, 20, 22]. Therefore, a theoretical model to understand the self-organization of dense nematic structures in actomyosin gels should consider a compressible and density-dependent fluid capturing the advective instability mentioned above. Furthermore, this model should acknowledge the active nature of the assembly of dense nematic structures, and permit extended isotropic phases commonly observed in the actin cortex, possibly coexisting with dense nematic phases [20, 31], rather than thermodynamically enforce high nematic order everywhere except at topological defects [36, 50–52].

Previous models for dry and dilute aligning active matter develop density patterns [37, 53, 54] but fail to capture the hydrodynamic interactions of actomyosin gels, whereas models for active nematic fluids either ignore density [38, 55–61], or account for the density of active particles suspended in an incompressible flow [34, 35, 62, 63], and therefore cannot describe the advective instability and self-reinforcing flows of actomyosin gels. Previous models describing the emergence of nematic patterns from uniform and isotropic states either ignore hydrodynamics [37], or the density and flow compressibility characteristic of actomyosin gels [38], and therefore result in very different instability mechanisms to those presented here. To model a thin layer of actomyosin gel, we summarize next a minimal theory for 2D density-dependent compressible active nematic fluids. In [64], we provide a systematic derivation of this theory based on a variational formalism of irreversible thermodynamics. This model can be understood as a density-dependent and compressible version of the active nematic theory presented in [57], or as a nematic generalization of the isotropic theory used in [29]. As elaborated in [64], it is possible to develop an alternative compressible and density-dependent active nematic theory based on the Beris-Edwards formalism [38, 51, 55, 56, 58, 59, 63].

In our model, the local state of the system is described by the areal density of cytoskeletal material $\rho(t, \mathbf{x})$ and by the network architecture given by the symmetric and traceless nematic tensor $\mathbf{q}(t, \mathbf{x})$, see **Fig. 1a**, both of which depend on time t and position $\mathbf{x}$. A more detailed model could consider separate densities for actin, myosin, and possibly other structural or regulatory proteins. Likewise, in principle, the orientational order of actin and myosin filaments could be described by separate nematic tensors. The nematic tensor can be expressed as $q_{ij} = S(n_i n_j - \delta_{ij}/2)$, where $\mathbf{n}$ is the average molecular alignment, $S = \sqrt{2q_{ij}q_{ij}}$ the degree of local alignment about $\mathbf{n}$ and δ_{ij} is the

identity. We denote by $\mathbf{v}(t, \mathbf{x})$ the velocity field of the gel. The rate-of-deformation tensor $d = \frac{1}{2}(\nabla \mathbf{v} + \nabla \mathbf{v}^T)$ measures the local rate of distortion of the fluid, whereas $w = \frac{1}{2}(\nabla \mathbf{v} - \nabla \mathbf{v}^T)$ measures its local spin, where ∇ is the nabla differential operator. The deviatoric part of the rate-of-deformation tensor is defined by $d_{ij}^{\text{dev}} = d_{ij} - (d_{kk}/2)\delta_{ij}$. The rate of change of $\mathbf{q}$ relative to a frame that translates and locally rotates with the flow generated by $\mathbf{v}$ is given by the Jaumann derivative $\dot{\mathbf{q}} = \partial \mathbf{q} / \partial t + \mathbf{v} \cdot \nabla \mathbf{q} - \mathbf{w} \cdot \mathbf{q} + \mathbf{q} \cdot \mathbf{w}$ [50].

Because we consider a bi-periodic domain Ω , we ignore other boundary conditions. Balance of cytoskeletal mass for a compressible fluid undergoing turnover takes the conventional form [29]

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) - D \Delta \rho + k_d(\rho - \rho_0) = 0, \quad (1)$$

where the second term models advection of cytoskeletal material by flow, the third term models diffusion with D being an effective diffusivity, and the last term models cytoskeletal turnover, where ρ_0 is the steady-state areal density and k_d is the depolymerization rate. Note that for a uniform, steady-state and quiescent gel, the first three terms vanish and $\rho(t, \mathbf{x}) = \rho_0$.

Force balance in the gel can be expressed as

$$\rho \gamma \mathbf{v} = \nabla \cdot \boldsymbol{\sigma}, \quad (2)$$

where $\gamma > 0$ models a viscous drag with the surroundings (e.g. the plasma membrane) and $\boldsymbol{\sigma} = \boldsymbol{\sigma}^{\text{nem}} + \boldsymbol{\sigma}^{\text{diss}} + \boldsymbol{\sigma}^{\text{act}}$ is the stress tensor in the gel, which in 2D has units of tension and which includes a contribution coming from the nematic free-energy, a dissipative contribution, and an active contribution.

The nematic stress $\boldsymbol{\sigma}^{\text{nem}}$ follows from a standard derivation adapted here to a density-dependent material. It derives from the free-energy $\delta = \int_{\Omega} \rho f(\mathbf{q}, \nabla \mathbf{q}) dS$, where

$$f(\mathbf{q}, \nabla \mathbf{q}) = \frac{1}{2} a S^2 + \frac{1}{8} b S^4 + \frac{1}{2} L |\nabla \mathbf{q}|^2 \quad (3)$$

is the classical Landau expansion of free-energy density per unit mass, with a and $b > 0$ susceptibility parameters, and $L > 0$ the Frank constant. When $a < 0$, the first term favors equilibrium nematic ordering, e.g. due to crowding in very dense gels of elongated filaments. Otherwise, the susceptibility terms entropically favor isotropic states with small S . In the actin cytoskeleton, this can model the random orientation of filaments as a result of the entropic fluctuations of filaments and their nucleators. The last term penalizes sharp gradients in the nematic field, which can result from the bending rigidity of actin filaments. A lengthy but standard calculation leads to the so-called molecular field [64]

$$h_{ij} = -\frac{\delta \mathcal{F}}{\delta q_{ij}} = -\rho (2a + b S^2) q_{ij} + L \nabla_k (\rho \nabla_k q_{ij}), \quad (4)$$

and to the explicit form of the nematic non-symmetric stress tensor

$$\sigma_{ij}^{\text{nem}} = -\rho \frac{\partial f}{\partial \nabla_j q_{ik}} \nabla_i q_{lk} + q_{ik} h_{jk} - q_{jk} h_{ik} = L [-\rho \nabla_i q_{kl} \nabla_j q_{kl} + q_{ik} \nabla_l (\rho \nabla_l q_{jk}) - q_{jk} \nabla_l (\rho \nabla_l q_{ik})]. \quad (5)$$

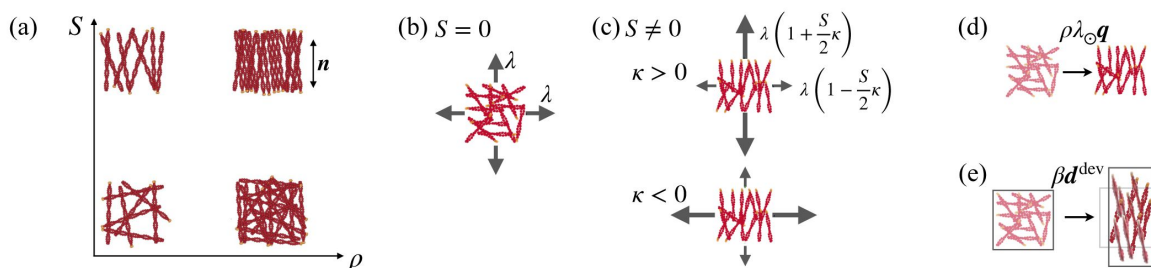


FIG. 1.

Key model ingredients

(a) The local state is defined by areal density ρ and by orientational order quantified by the nematic parameter S and by the nematic direction $\mathbf{n}$. (b) Isotropic active tension λ when the network is isotropic ($S = 0$), and (c) anisotropic tension when $S \neq 0$, controlled by κ . Active tension is positive (contractile) in all directions whenever $|\kappa| < 1$, but its deviatoric part is contractile when $\kappa > 0$ and extensile when $\kappa < 0$. Orientational order is driven by (d) active forces conjugate to nematic order and characterized by parameter $\lambda \odot$ and by (e) passive flow-induced alignment in the presence of deviatoric rate-of-deformation with coupling parameter β .

The dissipative part of the stress

$$\sigma_{ij}^{\text{diss}} = \rho [2\eta (d_{ij} + d_{kk}\delta_{ij}) + \beta \hat{q}_{ij}], \quad (6)$$

includes a viscous stress controlled by the gel viscosity parameter η [33] and a stress resulting from changes in the nematic field controlled by the coupling parameter $\eta < 0$ [7]. The term involving η can be understood as a stress in the gel arising from the drag induced by filaments as they reorient relative to the underlying hydrodynamic flow. Finally, we assume that the active stress resulting from the mechanical transduction of chemical power has an isotropic component and an anisotropic component oriented along the nematic tensor following

$$\sigma_{ij}^{\text{act}} = \rho (\lambda \delta_{ij} + \lambda_{\text{aniso}} q_{ij}) = \rho \lambda (\delta_{ij} + \kappa q_{ij}). \quad (7)$$

Activity parameter λ controls the isotropic tension and is contractile for $\lambda > 0$, as assumed here. The additional activity parameter λ_{aniso} , or equivalently $\kappa = \lambda_{\text{aniso}}/\lambda$, controls the deviatoric (traceless) component of active tension. This component is anisotropic and can be positive or negative parallel or perpendicular to the nematic direction depending on the sign of λ . When $|\lambda| < 1$, then the total active tension remains positive in all directions, with a larger magnitude parallel to the nematic direction when $\lambda > 0$ (contractile deviatoric component) and perpendicular to it when $\lambda < 0$ (extensile deviatoric component). We note that the isotropic component of active tension is meaningful here because our active gel is compressible. When order is low ($S \approx 0$), active tension is isotropic, **Fig. 1b**, whereas when order is high, active tension becomes anisotropic, **Fig. 1c**. We can interpret that active tension along the nematic direction reflects the sliding of antiparallel fibers driven by myosin motors, and whereas active tension perpendicular to it reflects the the out-of-equilibrium binding of bundling proteins or myosins [5, 65–71].

Balance of the generalized forces power-conjugate to q also includes viscous, elastic-nematic and active contributions, and takes the form

$$\eta_{\text{rot}} \hat{q} + \beta \mathbf{d}^{\text{dev}} - \frac{1}{\rho} \mathbf{h} - \rho \lambda_{\odot} \mathbf{q} = \mathbf{0}. \quad (8)$$

In this expression, η_{rot} is a nematic viscous coefficient. The second term models alignment induced by the rate of deformation of the flow, e.g. with compression/extension driving alignment perpendicular/parallel to the velocity gradient (**Fig. 1e**), as experimentally observed in [7]. This term involves the same coefficient β as the last term in Eq. (6) because of Onsager's reciprocity relations, and the entropy production inequality requires that $2\eta \eta_{\text{rot}} - \beta^2 \geq 0$ [64]. The third term is a thermodynamic force driven by F . In agreement with the observations that nematic ordering in actomyosin gels is actively driven, we assume $a > 0$, and therefore Eqs. (4, 8) show that this term tends to restore isotropy. The last term is an active generalized force controlled by activity parameter $\lambda_{\odot} \geq 0$ tending to further align filaments (**Fig. 1d**) [7]. This term is linear in ρ because in the expansion $\lambda_{\odot} + \rho \lambda_{\odot}$ the constant contribution $\lambda_{\odot}$ can be subsumed by the susceptibility parameter a [33]. Thus, the active term acts as a negative density-dependent susceptibility. When $c_0 = 2a - \rho_0 \lambda_{\odot} < 0$, the system can sustain a uniform quiescent state with $\rho(\mathbf{x}, t) = \rho_0$, $\mathbf{v}(\mathbf{x}, t) = \mathbf{0}$ and a non-zero nematic order parameter $S_0^2 = -c_0/b$. Even if $c_0 > 0$, and hence the uniform quiescent state is devoid of order, pattern formation can induce density variations such that $2a - \rho \lambda_{\odot}$ becomes negative locally and actively favors local nematic order. Physically, the term $-\rho \lambda_{\odot} \mathbf{q}$ implements the notion that the binding of a bundling protein, which drives active alignment, is more probable when two filaments are in close proximity and nearly aligned, a situation favored by high density and nematic order.

The nonlinearity of the coupled system of partial differential equations given by the balance laws Eqs. (1, 7, 8), along with the constitutive relations in Eqs. (4, 5, 6, 7), has different sources summarized next. The theory presents nonlinearities intrinsic to transport equations in the advective term of Eq. (1) and in the definition of the Jaumann derivative of the nematic tensor. Furthermore, nonlinearities in $\mathbf{q}$ in the constitutive relations results from the standard nematic free-energy adopted here. Our hypothesis that material properties in the gel are proportional to density and our thermodynamically-consistent derivation of the theory [64] result in further nonlinearities involving density in the constitutive relations. Finally, the nonlinearity involving density and the nematic field in the last term of Eq. (8) has been discussed in the previous paragraph.

Our theory has three active parameters, λ , κ and $\lambda_\odot$, all reflecting the conversion of chemical power into mechanical power in the network. The magnitude and the modes of chemomechanical transduction should depend on the molecular architecture of the network [72, 73], e.g. the stoichiometry of filaments, cross-linkers, and myosins, or the length distribution of filaments. Accordingly, we allow these parameters to vary independently.

By freezing an isotropic state, $S = 0$, our model reduces to an orientation-independent active gel model, which develops periodic patterns driven by self-reinforcing active flows sustained by diffusion and turnover [29]. In the present model, however, translational, orientational and density dynamics are intimately coupled through the terms involving β , $\lambda_\odot$, κ and L .

We readily identify the hydrodynamic length $\ell_s = \sqrt{\eta/\gamma}$, above which friction dominates over viscosity, the Damkolher length $\ell_D = \sqrt{D/\bar{k}_d}$ above which reactions dominate over diffusion, and the nematic length $\ell_q = \sqrt{L/|2a - \lambda_\odot \rho_0|}$. Non-dimensional analysis reveals a set of non-dimensional groups that control the system behavior, namely the non-dimensional turnover rate $\bar{k}_d = \ell_s^2/\ell_D^2$, the Frank constant $\bar{L} = L/(\eta D)$, the susceptibility parameters $\bar{a} = a/(\gamma D)$ and $\bar{b} = b/(\gamma D)$, the drag coefficients $\bar{\eta}_{\text{rot}} = \eta_{\text{rot}}/\eta$ and $\bar{\beta} = \beta/\eta$, the nematic activity coefficient $\bar{\lambda}_\odot = \rho_0 \lambda_\odot/(\gamma D)$, and the active tension parameters $\bar{\lambda} = \lambda/(\gamma D)$ and κ (Appendix B). The full list of material parameters for each figure given in Tables I and II and justified in Appendix D.

Results

Onset and nature of pattern formation

To examine the role of nematic order in the emergence of various actin architectures, we performed linear stability analysis of our model particularized to 1D, whose dynamical variables are velocity, density and nematic order, $v(x, t)$, $\rho(x, t)$, and $q(x, t)$, along x , where $q > 0$ (< 0) corresponds to a nematic orientation $\mathbf{n}$ parallel (perpendicular) to the x -axis (Appendix A). We first focused on the case $c_0 = 2a - \rho_0 \lambda_\odot > 0$ to examine the loss of stability of a uniform, isotropic, and quiescent steady state ($\rho(x, t) = \rho_0$, $v(x, t) = 0$, $q(x, t) = 0$) by increasing the master activity parameter λ and identifying the most unstable modes. This allowed us to determine a threshold activity for pattern formation and the wavelength of the emerging pattern (Appendix C). Since the exact evaluation of such quantities requires solving nonlinear equations, we derived explicit expansions in the limit of small L for the critical contractile activity

$$\lambda_{\text{crit}} \approx \lambda_{\text{crit},0} \left[1 - \frac{1}{2} \frac{\ell_s}{\ell_D} \left(1 + 2 \frac{\ell_s}{\ell_D} \right) \delta \right] + \mathcal{O}(\delta^2), \quad (9)$$

where $\lambda_{\text{crit},0} = (\sqrt{\gamma D} + 2\sqrt{\ell_s \eta})^2 = \gamma D(1 + 2\sqrt{\ell_s/\ell_D})^2$ and $\delta = \gamma D \kappa \beta / (2\eta c_0)$, and for the the corresponding wavenumber

$$\nu_{\text{crit}}^2 \approx \nu_{\text{crit},0}^2 \left[1 + \frac{1}{8} \left(1 + 2 \frac{\ell_s}{\ell_D} \right)^2 \delta \right] + \mathcal{O}(\delta^2), \quad (10)$$

where $\nu_{\text{crit},0}^2 = [k_d \gamma / (4\eta D)]^{1/2} = 1/(2\ell_s \ell_D)$.

When $\kappa = 0$ or $\beta = 0$, and hence $\delta = 0$, we recover the predictions of an active gel model not accounting for network architecture [29], $\lambda_{\text{crit}} = \lambda_{\text{crit},0}$ and $\lambda_{\text{crit}} = \lambda_{\text{crit},0}$. The expression for $\lambda_{\text{crit},0}$ shows that the instability takes place when activity is large enough to overcome the effect of diffusion and turnover, tending to uniformize density, and that of friction and viscosity, tending to suppress flows. Active tension anisotropy ($\kappa \neq 0$) and flow-induced alignment ($\beta < 0$) couple the instability of [29] to nematic order, changing the nature of pattern formation, see Eqs. (9,10) and the expression for δ . This leads to quantitative changes in critical tension and wavenumber, which can be very significant depending on the ratio of hydrodynamic and Damkohler lengths and on the strength and sign of nematic coupling. The nematic corrections increase as $c_0 \rightarrow 0$, close to the point where the uniform quiescent state develops spontaneous order. We thus studied separately the regime $0 < c_0 \ll 1$, finding analogous expansions for the critical tension and wavenumber in terms of $\delta = \gamma D \kappa \beta / (2\eta L)$. Interestingly, Eq. (9) shows that the activity threshold is reduced, and hence pattern formation facilitated, when $\kappa < 0$, i.e. when active tension is larger perpendicular to the nematic direction. Besides modifying critical tension and wavenumber, the present model predicts that the dynamical modes with self-reinforcing flows generate patterns where high density co-localizes with high nematic order.

To test the validity of this analysis and further understand the system beyond the onset of pattern formation, we performed fully non-linear finite element simulations in a periodic square 2D domain [64, 74], with a domain size chosen to be $8 \ell_{\text{pattern}}$ where $\ell_{\text{pattern}} = 2\pi/\nu_{\text{crit}}$ is the pattern lengthscale predicted by linear stability analysis. In these simulations, we increased the activity parameter A beyond the instability starting from a quiescent uniform state. We found that the linear stability analysis very accurately predicts the activity thresholds within two percent across a wide range of parameters. Furthermore, we found that the linear estimate for the wavenumber in Eq. (10) characterizes well the nonlinear patterns, as quantified in **Supp. Fig. 1** and illustrated visually by the density patterns in Figs. 2 and 4 showing the entire $8\ell_{\text{pattern}} \times 8\ell_{\text{pattern}}$ computational domain. In the nonlinear regime, the exponentially-growing instabilities eventually reach out-of-equilibrium quasi-steady-state patterns involving self-reinforcing flows towards regularly-spaced regions of high density surrounded by a low density matrix, as previously reported for various compressible and density-dependent active gel models [29, 39–43]. In the absence of nematic coupling ($\beta = \kappa = \lambda_{\odot} = 0$), these high density domains are droplets arranged in a regular hexagonal lattice with $S = 0$ throughout the domain as previously reported [40], **Movie 1**. In contrast, for a generic parameter set with finite β , κ and $\lambda_{\odot}$, high density domains adopt elongated configurations or bands where order is high, surrounded by a low density and low order matrix, **Fig. 2b** and **Movie 2**. Thus, our nematic active gel develops out-of-equilibrium localized nematic states starting from an isotropic state through a mechanochemical mechanism, which unlike those in [37, 38], exhibit localization of both density and nematic order, and hence resemble dense nematic structures embedded in a low-density isotropic actin cortex. Furthermore, the shape and internal architecture of dense and nematic phases are qualitatively modified by the nematic coupling.

Our simulations show that self-reinforcing flows develop along the direction of largest active tension, and consequently the pattern architecture depends on the sign of κ , **Fig. 2c**. For $\kappa < 0$, the system self-organizes into high-density and high-order bands, where nematic direction is parallel to their axis, in what we call *fibrillar pattern*, **Fig. 2b(I)**. Instead, for $\kappa > 0$ nematic order is perpendicular to the axis of the bands, in what we call *banded pattern*, **Fig. 2b(III)**. To

systematically study the effect of active tension anisotropy, we varied κ between -0.8 and 0.8 while keeping all other non-dimensional groups fixed and setting λ to be 1.3 times the critical activity. We defined the order parameter ω , [Fig. 2a](#), allowing us to distinguish between banded ($\omega > 0$) and fibrillar ($\omega < 0$) organizations. We found a sharp transition between fibrillar and banded regimes around $\kappa \approx 0$, during which elongated high-density and high-order domains fragment into nematic droplets or tactoids [\[75, 76\]](#), [Fig. 2b\(II\)](#).

Our results for $\kappa < 0$, leading to self-organized dense nematic fibrillar patterns from an isotropic low-density network, are in agreement with evidence suggesting that stress fibers can assemble from the actin cortex without the involvement of stress fiber precursors or actin polymerization at focal adhesions [\[20\]](#). They also agree with observations showing that actin bundles form a mechanical continuum with the surrounding sparse and isotropic cortex [\[31\]](#). Their morphology and patterning dynamics is strikingly reminiscent of actin microridges, formed at the apical surfaces of mucosal epithelial cells [\[77, 78\]](#). As discussed earlier, the condition $\kappa < 0$ implies that the deviatoric component of active tension is extensile. In agreement with widely studied incompressible extensile nematic systems, here the material is drawn perpendicular to the nematic direction, but rather than being expelled along the nematic direction as required from incompressibility [\[36\]](#), here it is recycled by disassembly and turnover ([Fig. 2c,d](#) left and Eq. (1)). Finally, we note the similarity in terms of density and nematic architecture between our fibrillar patterns and those emerging in other active systems through different mechanisms of self-organization, including polar motile filaments [\[79, 80\]](#) or mean-field models of dry mixtures of microtubules and motors [\[81\]](#).

Requirements for fibrillar and banded patterns

At linear order, our theory shows that the distinctly nematic self-organization requires both flow-induced alignment (β) and active tension anisotropy (κ), whereas no condition is required on nematic activity (λ_0). We performed further simulations to establish the requirements for fibrillar and banded active patterning in the nonlinear regime. We found that both banded and fibrillar patterns readily form for $\beta = 0$ and finite κ , yet a finite value of β enhances fibrillar formation, leading to longer and more stable dense bands, and hinders banded organization, [Movie 3](#). This behavior is expected since the velocity gradients of the self-reinforcing flows tend to align filaments parallel to high-density bands due to $\beta \mathbf{a}^{\text{dev}}$ in Eq. (8).

The active nematic coefficient λ_0 modifies the onset of pattern formation through c_0 according to linear stability analysis. Apart from this linear effect, it should also contribute to condensation of nematic order in high-density regions since it appears multiplied by p in Eq. (8). To examine this nonlinear effect, we perform simulations with $\lambda_0 = 0$ but keeping c_0 fixed. This leads to very different patterns without clear elongated structures and very mild nematic patterning, [Fig. 3a](#). Enhancing nematic patterning by considering the largest thermodynamically allowed value of $|\beta|$ leads to elongated structures for $\kappa < 0$, but rather than high-order co-localizing with high density, the nematic field develops domains with 90° angles between high- and low-density regions, [Fig. 3b](#), as observed in other active nematic systems [\[37\]](#). This architecture is enhanced by higher friction γ , [Fig. 3c](#). Thus, the architectures found for $\lambda_0 = 0$ and $\kappa < 0$ are distinct from the fibrillar pattern described previously. Similarly, rather than bands with order perpendicular to their axis, for $\lambda_0 = 0$, $\kappa > 0$ and high $|\beta|$ we found patterns of nematic asters (high-density droplets with radial nematic organization around them), [Fig. 3b,c](#).

Together, these results show that active tension anisotropy ($\kappa \neq 0$) and nematic activity ($\lambda_0 = 0$) are necessary and sufficient for nematic self-organization into fibrillar or banded patterns, with flow-induced alignment ($\beta < 0$) favoring fibrillar organization. Furthermore, they show that while Eqs. (9,10) accurately predict the activity threshold and the wavenumber of the nonlinear patterns, the linear stability analysis does not capture the architecture of the patterns, nor the key role of λ_0 , in the nonlinear regime.

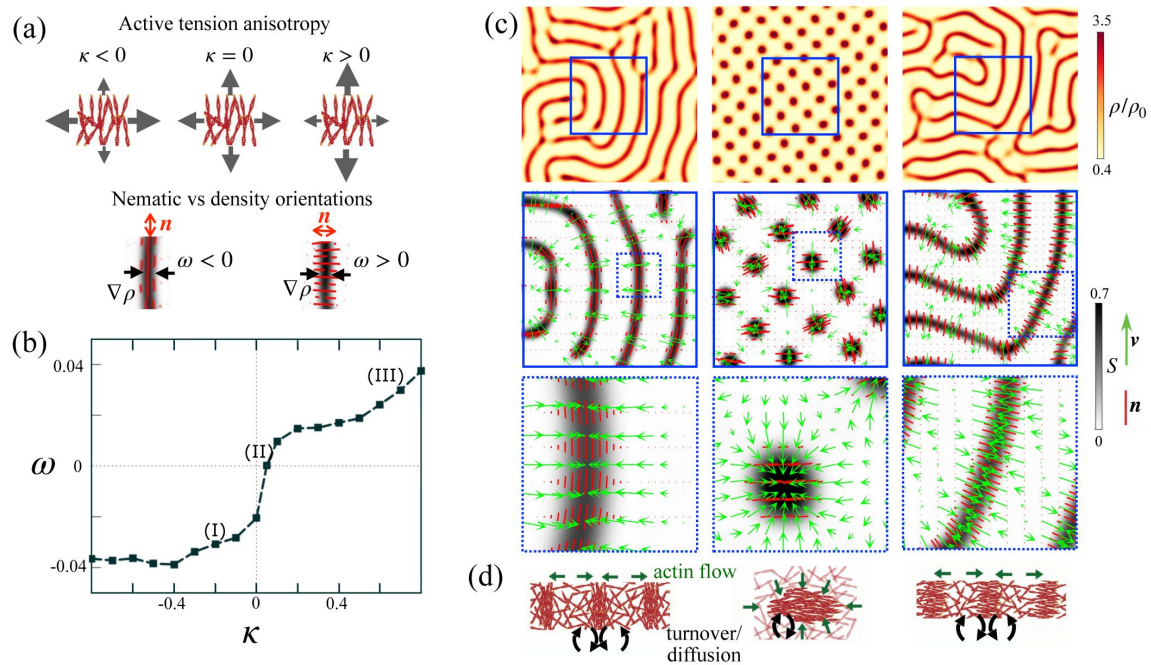


FIG. 2.

Active patterns coupling nematic order and density driven by self-reinforcing flows.

(a) Illustration of the dimensionless parameters characterizing active tension anisotropy (κ) and pattern architecture, quantified by the relative orientation of nematic order and high-density structures ($\omega = \ell_s^2 / (\rho_0^2 |A|) \int_A \nabla \rho \cdot \mathbf{q} \cdot \nabla \rho \, dS$). (b) Order parameter of pattern architecture ω as a function of active tension anisotropy κ obtained from nonlinear simulations, showing transition from states with nematic direction parallel to high-density structures ($\omega < 0$, fibrillar patterns) for $\kappa < 0$ to states with nematic direction perpendicular to high-density structures ($\omega > 0$, banded patterns with perpendicular nematic organization) for $\kappa > 0$. Because $|\kappa| < 1$, the active tension is always positive in all directions. (c) Map of density, nematic order S , nematic direction (red segments) and flow field (green arrows) for quasi-steady fibrillar (I) and banded (III) patterns, and for a transition pattern of high density droplets with high nematic order (II) corresponding to nearly isotropic active tension. (d) Illustration of the out-of-equilibrium quasi-steady states, maintained by self-reinforcing flows, diffusion and turnover.

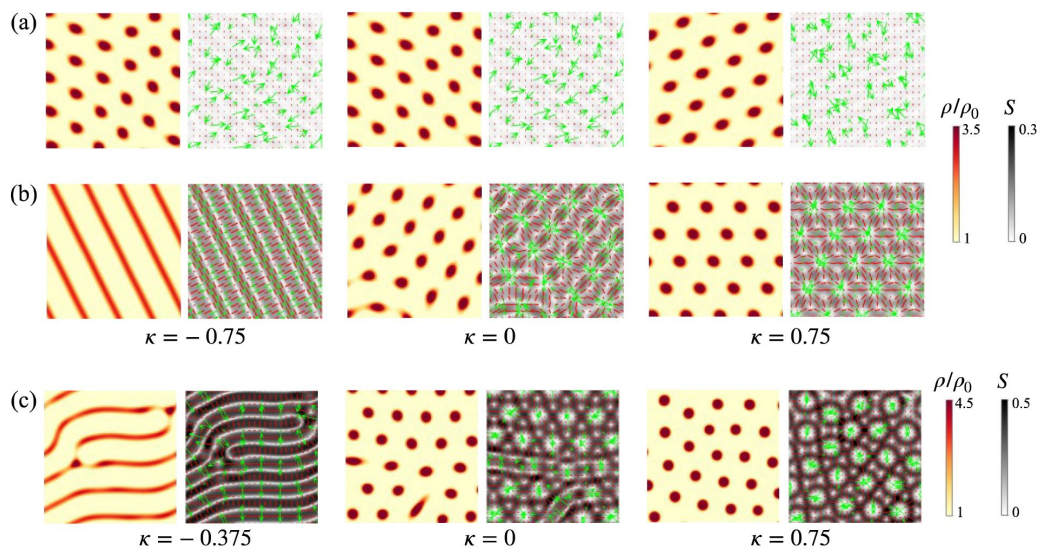


FIG. 3.

Pattern formation for a range of values of anisotropic active parameter κ in the limit $\lambda_{\odot} \rightarrow 0$.

(a) Pattern formation for material parameters used in [Fig. 2](#) except for $\lambda_{\odot} = 0$ while leaving c_0 unchanged. (b) Here, in addition to $\lambda_{\odot} = 0$, we set $\beta^2 = 2\eta\eta_{\text{rot}}$ to the largest value allowed by the entropy production inequality. (c) Same parameters as in (b), except for an increase in friction as detailed in [Table 1](#).

Morphological and dynamical diversity of self-organized fibrillar patterns

Given the morphological and dynamical diversity of nematic bundles in actin gels across cell types, geometric confinement, mechanical environment, or biological and pharmacological treatments [11, 26, 27, 82–84], we varied model parameters to examine the architectures predicted by our active gel model, focusing on $\kappa < 0$. Significant changes in the effective parameters of our active gel model are reasonable since the active mechanical properties of actomyosin gels strongly depend on micro-architecture both in reconstituted systems and in cells [70, 72].

With our default parameter set, bundle junctions and free ends are unfavorable and reorganize during pattern formation to annihilate as much as possible, [Movie 4\(I\)](#). However, because the initial state of the system is isotropic but fibrillar patterns are not, this coarsening process leads to frustrated labyrinth patterns with domains and defects, which depending on parameters can remain frozen in quasi-steady states as in [Fig. 4a\(I\)](#). We then asked the question of whether an orientational bias, which physically may be caused by cytoskeletal flows, boundaries or directed polymerization [22], could direct the pattern and result in fewer defects. We first slightly modified the system by including a small anisotropic strain-rate, according to which friction is computed relative to an elongating background. This directional bias is sufficient to produce well-oriented defect-free patterns aligned with the direction of elongation, [Fig. 4a\(II\)](#) and [Movie 4\(II\)](#). Alternatively, we considered c_0 to be slightly negative, leading to the uniform and nematic steady state $\rho(x,t) = \rho_0$, $v(x,t) = 0$ and $q(x,t) = q_0 = \pm \frac{1}{2} \sqrt{-c_0/b}$. The linear stability analysis around this state and further nonlinear simulations show that the essential phenomenology of Eqs. (9,10) and [Fig. 2](#) is not altered by the slight pre-existing order ([Appendix C](#)). Again, the weak pre-existing order provides sufficient bias to direct pattern orientation, [Fig. 4a\(III\)](#). Thus, our model indicates that an anisotropic bias can guide and anneal nematic fibrillar patterns, in agreement with remarkably oriented patterns of actin bundles in elongated cells [28], as a result of uniaxial cellular stretch [19, 78], or on anisotropically curved surfaces [21, 29].

At longer times, the nematic bundles in [Fig. 4a\(III\)](#) develop secondary active instabilities leading to coordinated bending, curvature amplification, defect nucleation and annihilation, [Movie 4\(III\)](#), in a behavior reminiscent of active extensile systems [34–36]. This possibility is far from obvious because, even though the deviatoric part of active tension is extensile since $\kappa < 0$, total active tension is contractile since $|\kappa| < 1$, [Fig. 1c](#). To systematically examine this point, we performed simulations at higher active tension anisotropies $\kappa = -0.8$, which we compared with our reference $\kappa = -0.2$, [Fig. 4b](#). While in our reference system bundles behave like contractile objects tending to straighten, for $\kappa = -0.8$ they behave like extensile objects enhancing curvature (see blue insets), which results in continuous defect nucleation as highly bent bundles destabilize and fragment, as well as defect annihilation as pairs of free ends merge to reorganize the network (see second inset in (b,II)), in a behavior akin to active turbulence [85]. See [Movie 5](#) for an illustration and for the slightly extensile case $\kappa = -0.5$. To further substantiate this contractile vs extensile behavior of the nematic structures emerging from a contractile active gel with extensile deviatoric behavior, we plotted the difference between the stress along the nematic direction and perpendicular to it, $\sigma_{||} - \sigma_{\perp}$. We note that for $\kappa < 0$ as required to obtain fibrillar patterns, the active component of this quantity is negative, $\sigma_{||}^{act} - \sigma_{\perp}^{act} < 0$, [Fig. 1c](#). In the case of bundles with contractile phenomenology, i.e. a tendency to straighten (b,I), we found that $\sigma_{||} - \sigma_{\perp}$ was positive on the dense bundles and nearly zero in between, indicating that bundles are contractile structures embedded in a largely isotropic matrix. Conversely, in bundles with extensile phenomenology (b,II), we found intricate stress patterns with regions of negative $\sigma_{||} - \sigma_{\perp}$. The extensile tension pattern for $\kappa = -0.8$ is more clearly appreciated in 1D simulations, [Supp. Fig. 2](#), where the dense nematic structures cannot fragment as a result of the bend-type instability. This figure also shows how the competition between active and viscous stresses dictates the emergent

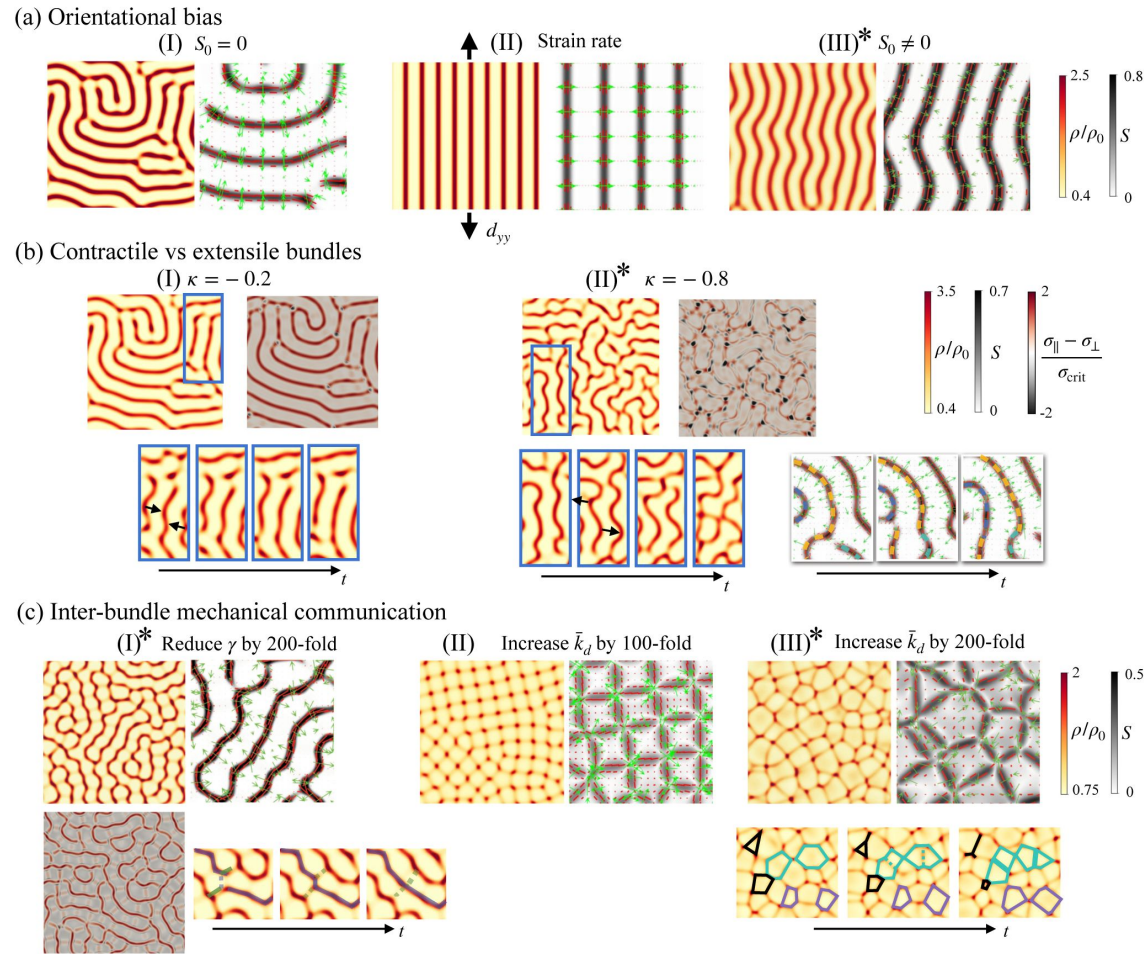


FIG. 4.

Control of nematic bundle pattern orientation, connectivity and dynamics.

(a) Effect of orientational bias. (I) A uniform isotropic gel self-organizes into a labyrinth pattern with defects. (II) A small background anisotropic strain-rate efficiently orients nematic bundles. (III) A slight initial network alignment ($S_0 = 0.05$) orients bundles, which later lose stability, bend, and generate/anneal defects. See [Movie 4](#). We recall that the nematic order parameter in the quiescent and uniform equilibrium state is $S_0 = 0$ if $c_0 \geq 0$ and $S_0 = \sqrt{-c_0/b}$ otherwise, with $c_0 = 2\alpha - \rho_0\lambda_\odot$. (b) Depending on active tension anisotropy, nematic bundles are contractile and straighten (I, $\kappa = -0.2$), leading to quasi-steady networks, or extensile and wrinkle (II, $\kappa = -0.8$), leading to bundle breaking and recombination, and persistently dynamic networks (III). See [Movie 5](#). The contractility or extensibility of the nematic bundles can be appreciated in the maps of the difference between the stress along the nematic direction ($\sigma_{||}$) and the stress perpendicular to it ($\sigma_{\perp}$), normalized by the constant $\sigma_{\text{crit}} = \lambda_{\text{crit}}\rho_0$. (c) Promoting mechanical interaction between bundles. (I) Dynamical pattern obtained by reducing friction, and thereby increasing $\bar{a}$, $\bar{b}$, $\bar{\lambda}_\odot$ and $\bar{k}_d$. Time sequence in the bottom indicates a typical reconfiguration event during which weak bundles (dashed) become strong ones (solid) and vice-versa. (II) Nearly static pattern obtained increasing $\bar{k}_d$, (III) which becomes highly dynamic by further increasing $\bar{k}_d$. Time sequence in the bottom indicates illustrate the collapse (black), expansion (purple) and splitting (green) of cells in the network. See [Movie 6](#). We indicate by * dynamical patterns exhibiting spatiotemporal chaos.

contractile vs extensile behavior of the nematic bundles. In the contractile case ($\kappa = -0.2$), even if $\sigma_{||}^{\text{act}} < \sigma_{\perp}^{\text{act}}$, the negative viscous stresses due to the self-reinforcing flows are significantly larger in magnitude perpendicular to the nematic direction, resulting in $\sigma_{||} - \sigma_{\perp}$. In summary, our theory predicts that a contractile nematic active gel can self-organize into fibrillar patterns with mesoscale bundles that are either contractile or extensile depending on the parameter regime. This complex interplay between the contractility vs extensibility of the base material and that of the mesoscale nematic pattern resonates with recent observations where tissues made of contractile cells behave collectively as extensile nematic systems [45, 86].

Focusing on contractile bundles, we then examined a different parameter regime known to trigger sustained pattern dynamics. For isotropic gels, previous work has shown that reducing friction triggers chaotic dynamics as the distance between high-density regions, $\ell_{\text{pattern}} = 2\pi/\nu_{\text{crit}}$, becomes comparable or smaller than the hydrodynamic length scale [29], thus enabling their mechanical interaction. In a model devoid of orientational order, reducing friction is equivalent to increasing $\bar{k}_d$. In our model, however, we can either reduce γ , which in non-dimensional terms means increasing $\bar{k}_d, \bar{a}, \bar{b}$ and $\bar{\lambda}_{\odot}$ in concert, or increase $\bar{k}_d$ while leaving all other non-dimensional parameters fixed. The first of these choices leads to dynamic and hierarchical networks with very dense and highly-aligned bundles, which coexist with perpendicular weak bundles with much lower density enrichment and ordering. These two families of bundles enclose cells of isotropic and sparse gel, **Fig. 4c(I)**. Junctions where two or more dense bundles meet are very unfavorable and short-lived, but junctions of two dense and a weak bundle are much more stable. Because bundles are mechanically coupled, the network actively reorganizes in events where dense bundles become weak bundles and vice-versa, inset and **Movie 6(I)**. We note that here $\sigma_{||} > \sigma_{\perp}$, **Fig. 4c(I)** and **Supp. Fig. 2d**, and hence the persistent dynamics are unrelated to the previously described behavior of extensile bundles.

The second choice to favor mechanical interaction of bundles, increasing $\bar{k}_d$ only, leads to very different networks with high-density aster-like clusters interconnected by straight actin bundles. Because now $\bar{\lambda}_{\odot}$ is not particularly large, order is low at the core of these clusters, enabling high-valence networks where four bundles often meet at one cluster. For $\bar{k}_d = 10$, the network is stable and nearly crystalline, **Fig. 4c(II)**, whereas for $\bar{k}_d = 20$, it becomes highly dynamical and pulsatile with frequent collapse of polygonal cells by fusion of neighboring actin clusters and their attached bundles (black polygons) and nucleation of new bundles within large low-density cells (dashed/solid green lines), **Fig. 4c(III)** and **Movie 6(III)**. This architecture and dynamics resemble those of early *C. elegans* embryos [87], adherent epithelial cells treated with epidermal growth factor [27] and mouse embryonic stem cells [83]. Recent active gel models accounting for RhoA signaling develop similar pulsatile behaviors in 2D, but do not predict the orientational order of the spatio-temporal patterns of the actomyosin cortex [88].

In summary, our theory maps how effective parameters of the actin gel control the active self-organization of a uniform and isotropic gel into a pattern of high-density nematic bundles embedded in a low-density isotropic matrix, including the activity threshold, the bundle spacing, orientation, connectivity and dynamics.

Microscopic origin of $\kappa < 0$ and $\lambda_{\odot} > 0$ through discrete network simulations

A central prediction of our model is that the self-organization of nematic bundles, the most prominent emerging organization in actin gels across cell types and length-scales, requires that active tension perpendicular to nematic orientation is larger than along this direction ($\kappa < 0$, see **Fig. 1c**), at least at the onset of pattern formation. This fact may seem counter-intuitive in that dense nematic bundles are associated with large contractile tension along their axis, but as discussed in **Fig. 4c(I)**, even if $\kappa < 0$ bundles can be contractile because of the interplay between

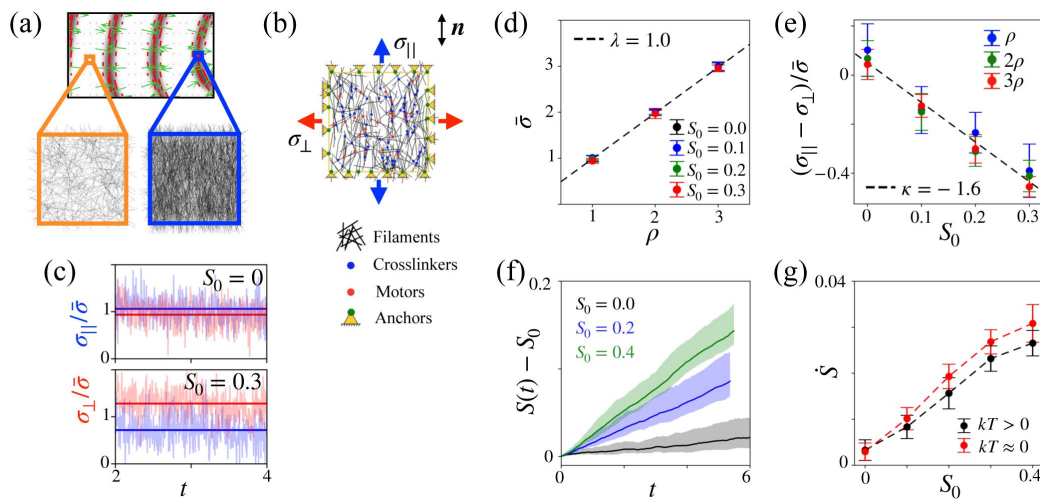


FIG. 5.

Assessment of activity parameters κ and λ_0 through discrete network simulations.

(a) Illustration of the computational domain of the discrete network as a uniform representative volume element of the gel. (b) Sketch of model ingredients and setup to compute tension along and perpendicular to the nematic direction. (c) Typical time-signal for parallel and perpendicular tensions following addition of crosslinkers and motors (translucent lines) along with time average (solid lines) for isotropic and anisotropic networks. Tension is normalized by mean tension $\bar{\sigma} = (\sigma_{\parallel} + \sigma_{\perp})/2$ computed from time-averages and time by actin turnover time. (d) Mean tension as a function of network density for several nematic parameters S_0 , where both quantities are normalized by their values for the lowest density. With this normalization, Eq. (11) predicts a linear dependence with slope $\lambda = 1$ (dashed line). Error bars span two standard deviations. (e) Deviatoric tension as a function of nematic order for different densities. The dashed line is a linear regression to simulation data. (f) Dynamics of nematic order in a periodic network following addition of crosslinkers and motors for three initial values of nematic order. (g) Rate of change of nematic order normalized by turnover rate as a function of initial nematic order at zero and finite temperature.

active and viscous stresses. Another prediction of our model is that the active assembly dense nematic bundles requires active alignment controlled by parameter $\lambda_{\odot} > 0$, **Fig. 3**. Being central to our conclusions, we then sought to examine the plausibility of the conditions $\kappa < 0$ and $\lambda_{\odot} > 0$ of our phenomenological theory by performing discrete network simulations using the open-source cytoskeleton simulation suite Cytosim [89].

Similarly to reconstituted actomyosin gels, discrete network simulations of the actin cytoskeleton tend to irreversibly collapse into clumps [70], although proper tuning of model parameters can lead to sustained contractile states [90]. However, to our knowledge, these models have not been able to reproduce the heterogeneous nonequilibrium contractile states involving sustained self-reinforcing flows underlying the pattern formation mechanism studied here. For this reason, rather than trying to reproduce the assembly and maintenance of patterns of dense nematic structures, we specifically focused on investigating whether the average behavior of the discrete network is compatible with the conditions $\kappa < 0$ and $\lambda_{\odot} > 0$ of our continuum theory. For this, we studied the buildup of tension and dynamics of nematic order in uniform representative volume elements of varying density and filament alignment after being brought out of equilibrium, **Fig. 5a**. Each of our simulated volume elements can represent a uniform system prior to pattern formation, or a material point in the actomyosin gel.

We performed 2D simulations in which semi-flexible filaments interact with crosslinkers and myosin motors, all of which undergo turnover and have a stoichiometry previously used to model the actin cytoskeleton [91], **Fig. 5b**. See **Appendix D** for a detailed description of the simulation protocol. Briefly, we modified Cytosim to account for average orientational order in the simulation box, which we evaluated as a sample average of orientations over the ensemble of segments composing the filaments. We further introduced a nematic energy penalty in the network allowing us to restrain average nematic order to a target value S_0 .

We first prepared a system consisting only of randomly oriented actin fibers, imposed the desired orientational order S_0 using the nematic penalty and equilibrated the system. In a first set of simulations, once S_0 was reached, we deactivated the nematic penalty and added crosslinkers and myosins, driving the system out-of-equilibrium. The free contraction of the system was prevented by the addition of anchors at the boundary of the representative volume element, which also allowed us to compute anchor forces and hence estimate the effective active tension along the nematic direction $\sigma_{||} = \sigma_{ij} \sigma_i \sigma_j$ and perpendicular to it, $\sigma_{\perp} = \sigma_{ij} \sigma_i m_j$ with $n_i m_i = 0$ and $m_i m_i = 1$, **Fig. 5b**.

Addition of crosslinkers and myosins leads to bundling of actin filaments at the microscale, **Movie 7**, distinct from the mesoscale fibrillar pattern formation emerging from the active gel model. It also leads to out-of-equilibrium tension as measured by the anchors. For isotropic networks ($S_0 = 0$), active tension is isotropic with $\sigma_{||} \approx \sigma_{\perp}$. For anisotropic networks, however, we found that tension becomes anisotropic with $\sigma_{\perp} > \sigma_{||}$, **Fig. 5c**.

We systematically characterized this behavior varying initial orientational order and network density. According to our active gel model, Eqs. (5, 6, 7), in the absence of nematic gradients and flow, the tension components $\sigma_{||}$ and $\sigma_{\perp}$ satisfy the following relations in terms of mean and deviatoric tensions

$$\bar{\sigma} = (\sigma_{||} + \sigma_{\perp})/2 = \lambda \rho \quad \text{and} \quad (\sigma_{||} - \sigma_{\perp})/\bar{\sigma} = \kappa S. \quad (11)$$

Remarkably, our discrete network simulations closely followed these relations, **Fig. 5d,e**, which allowed us to estimate $\kappa \approx -1.6$. We robustly found that $\kappa < 0$ for perturbations of selected parameters of the discrete network model as long as turnover rates of crosslinkers and myosins were relatively fast.

We then wondered if the discrete network simulations could provide evidence for the orientational activity parameter in our theory, $\rho\lambda_{\odot}$. For a uniform system with nematic order along a given direction and ignoring the susceptibility parameter b , **Eq. (8)** [↗](#) becomes

$$\eta_{\text{rot}}\dot{S} + (2a - \rho\lambda_{\odot})S + \mathcal{K}_S(S - S_0) = 0, \quad (12)$$

where η_{rot} is the viscous drag of the filaments in the discrete network simulations, $a > 0$ the entropic tendency of the model to return to isotropy, $\rho\lambda_{\odot}$ the active forcing of nematic order resulting from crosslinkers and motors, and the last term accounts for the effect of the nematic penalty with coefficient $\mathcal{K}_S$. As a first test of this model, we started from an isotropic and periodic network and tracked the athermal dynamics of S under the action of the nematic penalty in the absence of anchors, crosslinkers and myosins. For $\lambda_{\odot} = 0$ and $a = 0$, **Eq. (12)** [↗](#) predicts an exponential relaxation given by $S(t) = S_0(1 - e^{-\mathcal{K}_S t / \eta_{\text{rot}}})$, which very closely matched the simulation data for different values of S_0 and for a single fitting parameter η_{rot} , **Supp. Fig.3(III)** [↗](#) and **Movie 8** [↗](#). We then deactivated the nematic penalty and added crosslinkers and motors, but not anchors, to track unconstrained dynamics of nematic order starting from different values of S_0 . In agreement with the notion of an active force driving nematic order, we found that $S(t)$ monotonically increased, **Fig. 5f** [↗](#). More quantitatively, we tested the short-time prediction of **Eq. (12)** [↗](#), $\eta_{\text{rot}}\dot{S} = (\rho\lambda_{\odot} - 2a)S_0$, by plotting $\dot{S}$ as estimated from our simulations, as a function of S_0 , **Fig. 5g** [↗](#). We found a nearly linear relation with positive slope, hence providing evidence for an active generalized force driving order. In agreement with the theory, in the athermal limit, the tendency to actively increase order is faster as the entropic tendency to disorder is absent ($a = 0$).

In summary, discrete network cytoskeletal simulations provide a microscopic justification for two key ingredients of our active gel theory, namely that nematic order elicits (1) anisotropic active tensions, which can be larger perpendicular to the nematic direction ($\kappa < 0$), and (2) active generalized forces driving further ordering..

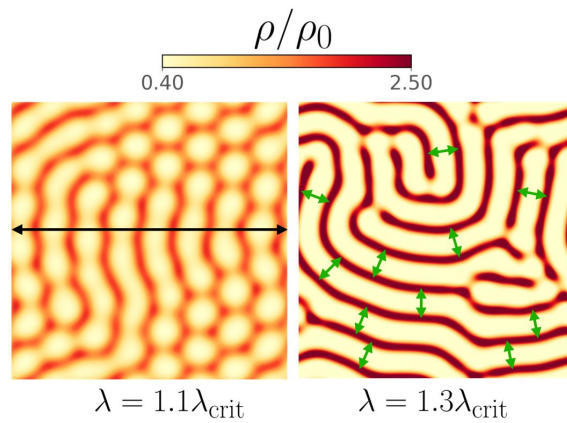
Conclusions

We have developed a theory for the active self-organization of initially uniform and isotropic actin gels into various localized dense nematic architectures embedded in an isotropic matrix of low density. This model predicts a variety of emergent patterns involving asters, tactoids, and bands with perpendicular nematic organization. More importantly, it identifies a wide parameter space where the active gel spontaneously develops patterns of dense nematic bundles, the most prominent nematic architecture across scales and cell types. Contrary to previous works on instabilities and pattern formation in active nematics, the mechanism proposed here relies on the advective instability and compressible self-reinforcing flows typical of actomyosin gels. We have characterized how the activity threshold, spacing, geometry, connectivity and dynamics these patterns depends on effective active gel parameters. Because these mesoscale parameters depend on the composition and dynamics of the network at the molecular scale, our results portray actin gels as responsive and reconfigurable active materials with an intrinsic ability to assemble patterns of nematic bundles that cells can finely regulate.

We have further shown that the spontaneous tendency of the gel to assemble bundle patterns can be directed via subtle cues. Thus, a combination of biochemical control of actin dynamics along with geometric, mechanical or biochemical guiding [92 [↗](#), 93 [↗](#)] may explain the emergence and context-dependent organization of regular patterns of bundle networks, from sub-cellular to organism scales [12 [↗](#), 14 [↗](#), 19 [↗](#), 21 [↗](#), 22 [↗](#), 25 [↗](#)–28 [↗](#), 30 [↗](#)]. Consistently, perturbations of myosin contractility have been shown to alter, disorder, or even prevent the formation patterns of parallel bundles in *C. elegans* [19 [↗](#)], whereas perturbations of actin polymerization in drosophila embryos impair the robust organization of actin bundle patterns at the cellular and organ scales

by disrupting orientation and spacing, but not the intrinsic tendency of the actomyosin cytoskeleton to form patterns of parallel bundles [21 [↗](#)]. In adherent cells, the patterns of dense nematic bundles presented here may act as precursors of stress fibers as the actomyosin cytoskeleton interacts with the focal adhesion machinery in a cellular domain with boundary conditions set by the polymerization velocity at the leading edge and the nucleus.

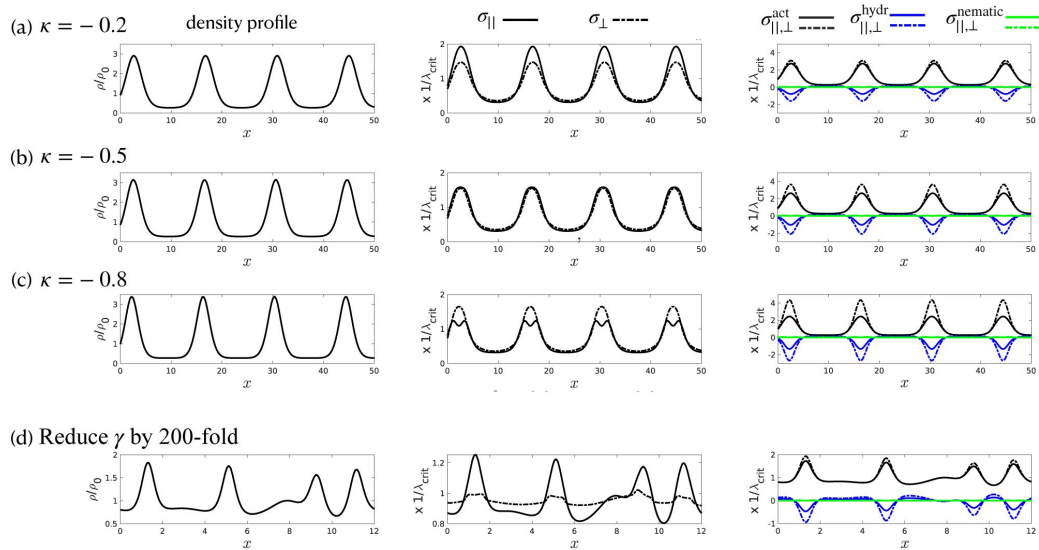
Our theory identifies two key requirements on activity parameters for the self-organization of patterns of nematic bundles, namely active tension anisotropy with larger tension perpendicular to the nematic direction and generalized active forces tending to increase nematic order. Complementary to our phenomenological theory, we have examined the plausibility of these conditions with discrete network simulations of homogeneous representative volume elements of different density and orientational order, which have confirmed our constitutive assumptions. We expect, however, that in a different parameter regime anisotropic tension may be larger along the nematic direction ($\kappa > 0$). For instance, once bundles are dense and maximally aligned, the ability of the active nematic gel to perform active tension perpendicular to the nematic direction may saturate, while myosin motors may contract the gel along the nematic direction more effectively. The regime studied here explains the initial assembly of dense nematic bundles, but not their maturation to become highly contractile or viscoelastic as demonstrated for stress fibers depending on different isoforms of nonmuscle myosin II or on actin regulators such as zyxin [94 [↗](#), 95 [↗](#)]. Our work thus suggests further experimental and computational work to establish a comprehensive mapping between molecular and mesoscale properties of the active gel, and how these properties control the emergent network architecture and mechanical properties.



Supp. Figure 1.

Lengthscale of the pattern.

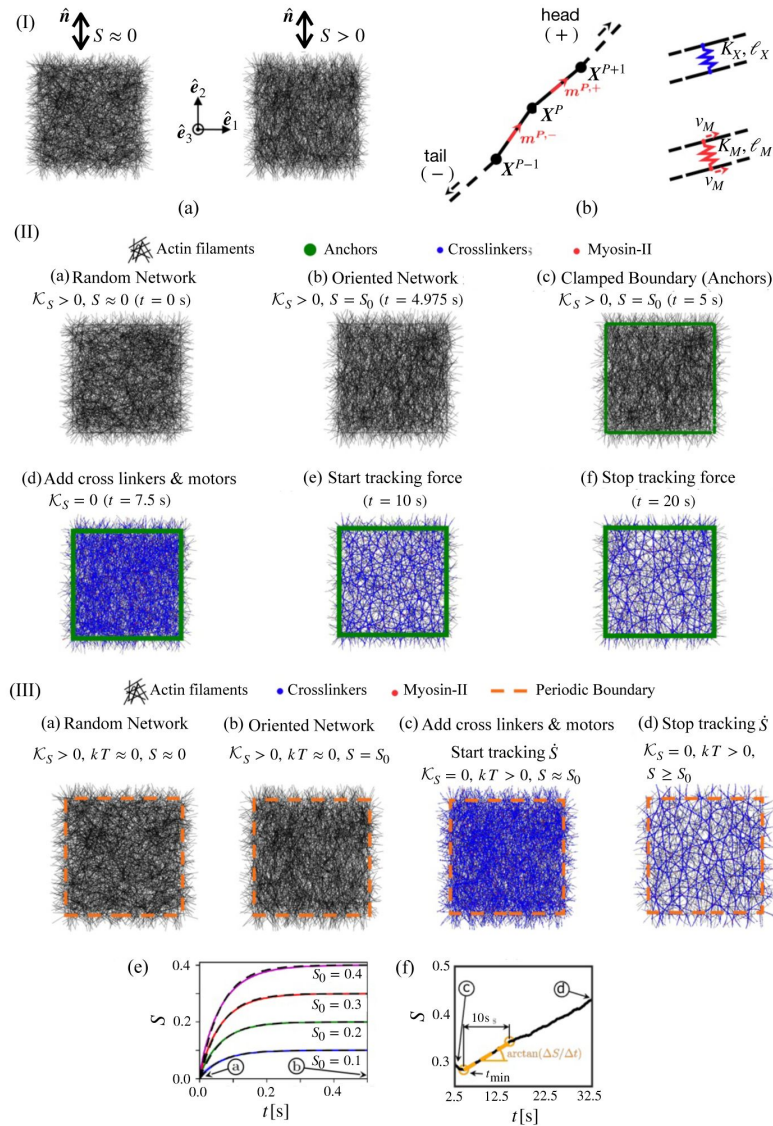
Illustration of the lengthscale of the pattern close to the onset of the instability (left) and deeper into the nonlinear regime (right). The linear stability estimate of the lengthscale of the pattern is $\ell_{\text{pattern}} = 2\pi/v_{\text{crit}}$, where v_{crit} is given by Eq. (10) in terms of material parameters. The side length of the square periodic computational domain is $8\ell_{\text{pattern}}$. Close to the onset (left), the pattern along the black arrow, with exactly 8 repeats, shows that the lengthscale in the nonlinear simulations closely follows the theoretical estimate. At activities 30% above the threshold (right), we sampled the typical separation between dense structures (green arrows) to approximate $\ell_{\text{pattern}}^{\text{sim}}$ from simulations, finding that $\ell_{\text{pattern}}^{\text{sim}}/\ell_{\text{pattern}} = 0.87 \pm 0.02$ (mean and standard deviation). We robustly found across the parameter space that the pattern lengthscale from nonlinear simulations 30% above the threshold was a little smaller than the theoretical estimate 'pattern, see density maps in Figs. 2 and 4.



Supp. Figure 2.

Tension distribution along ($\sigma_{||}$) and perpendicular to ($\sigma_{\perp}$) the dense nematic bundles.

The left column shows the density distribution, the middle column the total tension along (solid) and perpendicular (dashed) to nematic bundles, and the right column the different contributions to the total tension dominated by the active (black) and the viscous (blue) components. In all cases, we consider for convenience fully nonlinear simulations in 1D to easily define the orthogonal directions relative to the self-organized pattern. (a) to (c) show patterns obtained for negative tension anisotropy κ of increasing magnitude and correspond to **Fig. 3b** in the main text, whereas (d) shows a chaotic pattern resulting from a large hydrodynamic length and corresponds to **Fig. 3c(I)** in the main text. The model parameters used in the plots are the same as in **Figs. 2** and **3** and are described in **Tables I** and **II**



Supp. Figure 3.

Discrete network simulations.

(I) Microstructural modeling approach using cytosim. (a) The nematic ordering of the network, S , measured with respect to a director $\hat{n}$, is controlled by a system-wide restraining energy. (b) Actin filaments are represented by connected points kept at a fixed distance and have finite bending rigidity κ_A . Crosslinkers are Hookean springs with stiffness ℓ_X and resting length K_X . Myosin motors are Hookean springs with stiffness ℓ_M and resting length ℓ_M ; their ends can walk on filaments at a speed v_M , which is affected by force application. (II) Simulation protocol to quantify active tension anisotropy. (a) Initial fiber seeding without orientational bias. (b) Athermal imposition of desired order parameter S_0 using a restraining potential. (c) Introduction of boundary anchors. (d) Deactivation of orientational restraining potential and addition of crosslinkers and myosin motors, which drive the system out-of-equilibrium. Reactions at anchors allow us to track tensions at the boundary, (e,f). (III) Protocol to quantify the orientational activity parameter $\rho\lambda_O$. (a) Initial fiber seeding without orientational bias in a periodic box. (b) Athermal imposition of desired order parameter S_0 by restraining potential. (c,d) Activation of temperature, deactivation of orientational restraining potential, and addition of crosslinkers and myosin motors. The dynamics of orientational order of the system driven out-of-equilibrium are then tracked during a time period (c,d). (e) Estimation of the model parameter η_{rot} which characterizes the athermal dynamics of network reorientation in the presence of a restraining potential of stiffness K_S ; note that a single value of η_{rot} fits the dynamics for all considered S_0 . (f) The rate of change of nematic order, $\dot{S}$, is estimated from a 10 s time interval.

Table I

Model parameters used in figures.

Parameters	Fig. 2	Fig. 3a	Fig. 3b	Fig. 3c(I)	Fig. 3c(II,III)	Fig. 3a	Fig. 3b	Fig. 3c
$\bar{k}_d$	0.1	0.1	0.1	20	10,20	0.1	0.1	0.01
$\bar{L}$	1	1	1	1	1	1	1	1
$\bar{c}_0$	4	4,4,-0.05	4	200	4	4	4	0.4
$\bar{b}$	20	20	20	4000	20	20	20	2
$\bar{\eta}_{\text{rot}}$	1	1	1	1	1	1	1	1
$\bar{\beta}$	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	$-\sqrt{2}$	$-\sqrt{2}$
$\bar{\lambda}_{\odot}$	6	6,6,10.05	6	1800	6	0	0	0
$\bar{\lambda}$	$1.3\bar{\lambda}_{\text{crit}}$	$1.3\bar{\lambda}_{\text{crit}}$	$1.3\bar{\lambda}_{\text{crit}}$	$1.3\bar{\lambda}_{\text{crit}}$	$1.3\bar{\lambda}_{\text{crit}}$	$1.3\bar{\lambda}_{\text{crit}}$	$1.3\bar{\lambda}_{\text{crit}}$	$1.3\bar{\lambda}_{\text{crit}}$
κ	$[-0.8,0.8]$	-0.2	-0.2,-0.8	-0.2	-0.2	-1.5,-0.75,0,0.75	-1.5,-0.75,0,0.75	-0.75,-0.375,0,0.75
Domain Size $\times \nu_{\text{crit}}/(2\pi)$	8	8	8	8	8	8	8	8

Table II

Model parameters used in movies that do not directly reproduce figures.

Parameters	Movie 1	Movie 3 (top)	Movie 3 (bottom)	Movie 5
$\bar{k}_d$	0.01	0.1	0.1	0.1
$\bar{L}$	1	1	1	1
$\bar{c}_0$	0.4	4	4	4
$\bar{b}$	2	20	20	20
$\bar{\eta}_{\text{rot}}$	0	1	1	1
$\bar{\beta}$	0	-0.2,0	$-\sqrt{2},-0.2$	-0.2
$\bar{\lambda}_{\odot}$	0	6	6	6
$\bar{\lambda}$	$1.3\bar{\lambda}_{\text{crit}}$	$1.3\bar{\lambda}_{\text{crit}}$	$1.3\bar{\lambda}_{\text{crit}}$	$1.3\bar{\lambda}_{\text{crit}}$
κ	0	-0.2	0.2	-0.2,-0.5,-0.8
Domain Size $\times \nu_{\text{crit}}/(2\pi)$	8	8	8	8

Table III

Global parameters adopted in this study and in previous microstructural models that used cytosim.

Reference	Geometry and size [μm]	kT [pN μm]	Time step [s]	ν [Pas]
Present work	Square: $\ell = 5$	0 or 0.0042	0.005	1
MM1	Rectangle: 2×0.2	0.0042	0.001	1
[91] MM4	Rectangle: 9.424×1	0.0042	0.001	1
MM3 & MM5	Circle: $R = 1.5$	0.0042	0.001	1
Cortex simulations in [102]	Square: $L = 8$	0.0042	0.002	0.18
[103]	Circle: $R = 10$	0.0042	0.01 or 0.1	0.3
Model with turnover in [99]	Square: $L = 16$	0.0042	0.001	0.1
[104]	Circle: $R = 15$	0.0042	0.002	1
[105]	Circle: $R = 10$	0.0042	0.001	0.1
[70]	Ring: $R = 4.5, t \approx 0.5$	0.0042	0.01	0.18

Table IV

Actin filament parameters adopted in this study and in previous microstructural models that used cytosim.

Reference	ℓ_A [μm]	ρ_A [$\mu\text{m}/\mu\text{m}^2$]	s_A [μm]	κ_A [$\text{pN}\mu\text{m}^2$]	r_A [$\%/s$]	
Present work	1.3	78, 156, or 234	0.2	0.1	20	
	MM1	2	1800	0.05 – 0.1	0.06	0
Ref. [91]	MM3	1.3 ± 0.3	50.9 – 81.5	0.05 – 0.1	0.06	0
	MM4	1.3 ± 0.3	38.2 – 61.1	0.05 – 0.1	0.06	0
	MM5	1.3 ± 0.3	50.9 – 81.5	0.05 – 0.1	0.06	0
Cortex simulations in Ref. [102]	0.1 – 4.0	5.5 – 218.8	0.1 – 0.4	0.075	0	
Ref. [103]	1.5	23.9	0.15	0.07	0	
Model with turnover in Ref. [99]	5	27.3	0.1 – 0.2	0.075	1.1 – 18.3	
Ref. [104]	1.3 ± 0.3	3.4 – 5.4	0.1	0.06	0	
Ref. [105]	2.2	35	—	0.05	0	
Ref. [70]	0.95 – 1.75	117.2 – 154.2	—	0.042 – 0.063	0	

Table V

Myosin motor parameters adopted in this study and in previous microstructural models that used cytosim.

Reference	ρ_M [$1/\mu\text{m actin}$]	K_M [$\text{pN}/\mu\text{m}$]	ℓ_M [μm]	r_M^b [$1/s$]	d_M^b [μm]	$r_M^{u,0}$ [$1/s$]	f_M^u [pN]	f_M^{stall} [pN]	v_M^{max} [$\mu\text{m}/s$]
Present work	0.8	250	0	50	0.02	50	∞	6	0.3
[91]	0.5	100	0.3	0.2 – 3.6	0.06 – 0.12	0.8 – 1.71	5	3.85 – 15	0.137 – 0.6
Cortex simulations	0.6 – 1.0	100	0.3	0.2 – 3.6	0.06 – 0.12	0.8 – 1.71	5	3.85 – 15	0.137 – 0.6
[102]	2 – 64	100	0	10	0.01	0.5	∞	4	2
Ref. [103]	2.7	250	0.01	10	0.01	0.1	3	6	0.02
Model with turnover	3.2	500	0	10	0.01	0.3	∞	6	0.2
[99]									
[104]	0.5 – 0.8	1'400	0.32	0.2	0.33	0.3	3.85	24.5	0.1
[105]	0 or 5.8	250	0	10	0.01	0.5	∞	6	0.5
[70]	0.3 – 0.4	100	0.03	5	0.05	0	3.65	2	0.3

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Additional information

Author Contributions

WM, ATS and MA conceived the study and developed the active gel theory. WM, GV and ATS developed and implemented numerical methods. WM performed and analyzed active gel simulations. WM and MP performed and analyzed discrete network simulations. MP developed the code and simulation protocols for discrete network simulations. MDC contributed to linear stability analysis. WM and MA wrote the manuscript. ATS and MA supervised the study.

Additional files

Movie 1: Pattern formation in an active gel model not accounting for nematic order. [🔗](#)

Movie 2: Pattern formation for different values of the tension anisotropy coefficient κ , leading to (I) fibrillar patterns for $\kappa \approx 0$, (II) tactoids for $\kappa \ll 0$, and (III) banded patterns for $\kappa > 0$. This movie corresponds to Fig. 2b. [🔗](#)

Movie 3: Effect of flow-alignment coupling coefficient β on pattern formation for positive and negative tension anisotropy coefficient κ . Flow-alignment favors fibrillar patterns and disfavors banded patterns. [🔗](#)

Movie 4: Effect of orientational bias on fibrillar patterns. This movie corresponds to Fig. 3a. [🔗](#)

Movie 5: Contractile vs extensile bundles as the magnitude of κ increases, leading to active turbulence of fibrillar patterns. This movie correspond to Fig. 3b. [🔗](#)

Movie 6: Changes in network geometry and dynamics following enhanced inter-bundle mechanical communication. This movie correspond to Fig. 3c. [🔗](#)

Movie 7: Illustration of simulation protocol to estimate tension anisotropy from discrete network simulations for an isotropic (left) and anisotropic (right) representative volume elements. The phases are (1) alignment of filaments to reach the desired nematic order, (2) addition of boundary anchors (green dots) to prevent network collapse and to measure tension, (3) addition of crosslinkers and motors (blue and red dots) to bring the system out-of-equilibrium while tracking tension along horizontal and vertical directions. [🔗](#)

Movie 8: Illustration of simulation protocol to estimate the generalized active tension conjugate to nematic order for an isotropic (left) and anisotropic (right) representative volume elements with periodic boundary conditions. The phases are (1) alignment of filaments to reach desired nematic order, (2) addition of crosslinkers and motors (blue and red dots) to bring the system out-of-equilibrium while tracking the average nematic order of the representative volume element. [↗](#)

Author response video 1 [↗](#)

Appendix A: 1D reduced model

To perform the linear stability analysis, we reduce the theoretical model to 1D by considering the flow velocity $v(x, t)$ and the density field as $\rho(x, t)$ along the x -axis. The nematic order tensor is defined as $q_{ij} = S(n_i n_j - \delta_{ij}/2)$, where $\mathbf{n}$ is the local average filament orientation and S is the local degree of alignment. Since it is traceless and symmetric, in general it can be represented by 2 independent degrees of freedom, $q_{11} = -q_{22} = q_1$ and $q_{12} = q_{21} = q_2$. In the 1D model we assume that $\mathbf{n}$ is either along or perpendicular to x , and hence $q_{12} = q_{21} = 0$. We are thus left with one independent degree of freedom $q_{11} = -q_{22} = q$. Positive (negative) q represents alignment along (perpendicular to) the x -axis. The Jaumann derivative of the nematic order parameter reduces in 1D to

$$\hat{q} = \frac{\partial q}{\partial t} + v \frac{\partial q}{\partial x}. \quad (\text{A1})$$

Particularizing the general equations given in the main text, we present next the 1D governing equations pertinent to the linear stability analysis. Mass conservation reads

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x} (\rho v) - D \frac{\partial^2 \rho}{\partial x^2} + k_d (\rho - \rho_0) = 0, \quad (\text{A2})$$

balance of linear momentum of the active gel reduces to

$$\rho \gamma v = \frac{\partial \sigma}{\partial x}, \quad (\text{A3})$$

where stress along x , σ , is given by

$$\sigma = \rho \left[4\eta \frac{\partial v}{\partial x} + \beta \hat{q} + \lambda (1 + \kappa q) - 2L \left(\frac{\partial q}{\partial x} \right)^2 \right]. \quad (\text{A4})$$

The evolution of nematic order q is governed in the 1D setting by

$$\eta_{\text{rot}} \hat{q} + \frac{\beta}{2} \frac{\partial v}{\partial x} + (2a - \rho \lambda_{\odot} + 4bq^2) q - L \frac{\partial^2 q}{\partial x^2} - \frac{L}{\rho} \frac{\partial q}{\partial x} \frac{\partial \rho}{\partial x} = 0. \quad (\text{A5})$$

For $c_0 = 2a - \rho_0 \lambda_{\odot} \geq 0$, the uniform steady state of the system is given by $\rho(x, t) = \rho_0$, $v(x, t) = 0$ and $q(x, t) = 0$. For $c_0 < 0$, the uniform steady state has spontaneous alignment given by $q(x, t) = q_0 = \pm \frac{1}{2} \sqrt{-c_0/b}$. Finally, we note that, for the dissipation to be positive, the material parameters need to satisfy an entropy production inequality $4\eta\eta_{\text{rot}} - \beta^2 \geq 0$ in the reduced one-dimensional model and $2\eta\eta_{\text{rot}} - \beta^2 \geq 0$ in the two-dimensional model [64].

Appendix B: Non-dimensionalization

We present next a non-dimensionalization of the governing Eqs. (A2–A5). We choose the hydrodynamical lengthscale $\ell_s = \sqrt{\eta/\gamma}$, above (below) which friction (viscosity) is the dominant dissipative mechanism in the active gel, and the time-scale of diffusion over this length-scale, $t_0 = \ell_s^2/D = \eta/(\gamma D)$. Using these characteristic scales, we non-dimensionalize space and time as $\bar{x} = x/\ell_s$ and $\bar{t} = t/t_0$. We chose ρ_0 as a reference density and thus $\bar{\rho} = \rho/\rho_0$. Here and elsewhere, overbar denotes non-dimensional quantities.

The dimensionless balance of mass reads

$$\frac{\partial \bar{\rho}}{\partial \bar{t}} + \bar{v} \frac{\partial \bar{\rho}}{\partial \bar{x}} + \bar{\rho} \frac{\partial \bar{v}}{\partial \bar{x}} - \frac{\partial^2 \bar{\rho}}{\partial \bar{x}^2} + \bar{k}_d(\bar{\rho} - 1) = 0, \quad (\text{B1})$$

where $\bar{k}_d = k_d t_0$. The dimensionless balance of linear momentum reads

$$\bar{\rho} \bar{v} = \frac{\partial \bar{\sigma}}{\partial \bar{x}}, \quad (\text{B2})$$

where $\bar{\sigma} = \sigma/\sigma_0$ and the characteristic stress is given by $\sigma_0 = \rho_0 \gamma D$. The non-dimensional constitutive relation for the stress is

$$\bar{\sigma} = \bar{\rho} \left[4 \frac{\partial \bar{v}}{\partial \bar{x}} + \bar{\beta} \bar{q} + \bar{\lambda} (1 + \kappa q) - 2 \bar{L} \left(\frac{\partial q}{\partial \bar{x}} \right)^2 \right], \quad (\text{B3})$$

where $\bar{L} = L/(\gamma D \ell_s^2)$, $\bar{\beta} = \beta/\eta$, and $\bar{\lambda} = \lambda/(\gamma D)$. We note that $\kappa = \kappa_{\text{aniso}}/\kappa$ and q are already nondimensional. Finally, the dimensionless equation of generalized force conjugate to nematic order is given by

$$\bar{\eta}_{\text{rot}} \bar{q} + \frac{\bar{\beta}}{2} \frac{\partial \bar{v}}{\partial \bar{x}} + (2\bar{a} - \bar{\rho} \bar{\lambda}_{\odot} + 4\bar{b} q^2) q - \bar{L} \frac{\partial^2 q}{\partial \bar{x}^2} - \frac{\bar{L}}{\bar{\rho}} \frac{\partial q}{\partial \bar{x}} \frac{\partial \bar{\rho}}{\partial \bar{x}} = 0. \quad (\text{B4})$$

where $\bar{\eta}_{\text{rot}} = \eta_{\text{rot}}/\eta$, $\bar{a} = a/(\gamma D)$, $\bar{b} = b/(\gamma D)$ and $\bar{\lambda}_{\odot} = \lambda_{\odot} \rho_0/(\gamma D)$. Inspection of the Eq. (B4) reveals the characteristic nematic length-scale $\ell_q^2 = L/|2a - \lambda_{\odot} \rho_0|$ and the nematic relaxation time-scale $t_q = (\ell_q^2 \eta_{\text{rot}})/L = \eta_{\text{rot}}/|2a - \lambda_{\odot} \rho_0|$.

Appendix C: Linear stability analysis

We perform next the linear stability analysis on the 1D model given by the dimensionless Eqs. (B1–B4). For the sake of simplicity of notation, we omit overbars for the dimensionless quantities. We assess the stability of the homogeneous state given by $v(x, t) = 0$, $q(x, t) = q_0$ and $\rho(x, t) = \rho_0$ by examining the fate of infinitesimal perturbations $v(x, t) = \delta v(x, t)$, $q(x, t) = q_0 + \delta q(x, t)$ and $\rho(x, t) = \rho_0 + \delta \rho(x, t)$, where $|\delta v| \sim |\delta q| \sim |\delta \rho| \ll 1$. The control parameter in our stability analysis is the activity parameter κ .

Inserting the infinitesimally perturbed homogeneous states into the governing Eqs. (B1–B4) and ignoring terms that are quadratic in perturbations, the resulting balance of linear momentum equation is

$$\delta v = 4 \frac{\partial^2 \delta v}{\partial x^2} + \beta \frac{\partial^2 \delta q}{\partial t \partial x} + \lambda(1 + \kappa q_0) \frac{\partial \delta \rho}{\partial x} + \lambda \kappa \frac{\partial \delta q}{\partial x}. \quad (\text{C1})$$

The linearized form of Eq. (B4) is

$$\eta_{\text{rot}} \frac{\partial \delta q}{\partial t} + \frac{\beta}{2} \frac{\partial \delta v}{\partial x} + \underbrace{[2a - \rho_0 \lambda_\odot + 4bq_0^2 + 8bq_0^2]}_G \delta q - \lambda_\odot q_0 \delta \rho - L \frac{\partial^2 \delta q}{\partial x^2} = 0. \quad (\text{C2})$$

If $c_0 = 2a - \rho_0 \kappa_\odot > 0$, then $q_0 = 0$ and $G = c_0$. If $c_0 = 2a - \rho_0 \kappa_\odot < 0$, then $q_0^2 = -c_0/(4b)$ and $G = 8bq_0^2$. The linearized balance of mass reads

$$\frac{\partial \delta \rho}{\partial t} + \frac{\partial \delta v}{\partial x} - \frac{\partial^2 \delta \rho}{\partial x^2} + k_d \delta \rho = 0. \quad (\text{C3})$$

We decompose perturbations into a sum of Fourier terms, i.e., $\delta v = v' e^{at+ixv}$, $\delta \rho = \rho' e^{at+ixv}$ and $\delta q = q' e^{at+ixv}$, where v' , q' and ρ' are the amplitudes of the perturbations, a is the growth rate and v is the wavenumber.

Substituting the Fourier representation in Eq. (C1), we obtain

$$v'(1 + 4\nu^2) = q' i\nu(\beta\alpha + \lambda\kappa) + \rho' i\nu\lambda(1 + \kappa q_0). \quad (\text{C4})$$

Likewise, Eq. (C3) becomes

$$v' = \frac{\nu^2 + k_d + \alpha}{\nu} i\rho'. \quad (\text{C5})$$

Substituting this equation into the Fourier representation of Eq. (C2) we obtain

$$q' = \frac{\beta(\alpha + \nu^2 + k_d) + 2\lambda_\odot q_0}{2\eta_{\text{rot}}\alpha + 2G + 2L\nu^2} \rho'. \quad (\text{C6})$$

Finally, combining the three equations above, we obtain the dispersion equation, which, given the material parameters, relates wavenumber ν to growth rate a and takes the form

$$A(\nu)\alpha^2 + B(\nu)\alpha + C(\nu) = 0, \quad (\text{C7})$$

where

$$\begin{aligned} A(\nu) &= 2\eta_{\text{rot}}(4 + \nu^{-2}) - \beta^2, \\ B(\nu) &= (4 + \nu^{-2})(2G + 2L\nu^2) + 2\eta_{\text{rot}}[(4 + \nu^{-2})(\nu^2 + k_d) - \lambda(1 + \kappa q_0)] - \\ &\quad \beta[\beta(\nu^2 + k_d) + 2\lambda_\odot q_0 + \lambda\kappa], \\ C(\nu) &= [(4 + \nu^{-2})(\nu^2 + k_d) - \lambda(1 + \kappa q_0)](2G + 2L\nu^2) - \lambda\kappa[\beta(\nu^2 + k_d) + 2\lambda_\odot q_0], \end{aligned}$$

The roots of Eq. (C7) are given by

$\alpha(\nu) = \left[-B(\nu) \pm \sqrt{B(\nu)^2 - 4A(\nu)C(\nu)} \right] / (2A(\nu))$. The system is linearly stable if perturbations decay, i.e. Real (α) < 0, marginally stable if Real (α) = 0, and unstable if Real (α) > 0. We found that for all reasonable material parameters $\alpha(\nu)$ is real, and hence the critical condition is given by $\alpha = 0$, which recalling Eq. (C7) implies that $C(\nu) = 0$. From this condition, we can express the activity parameter as

$$\lambda = \frac{(\nu^{-2} + 4)(\nu^2 + k_d)(2G + 2L\nu^2)}{(1 + \kappa q_0)(2G + 2L\nu^2) + \kappa [\beta(\nu^2 + k_d) + 2\lambda_\odot q_0]} \quad (\text{C8})$$

To find the critical wavenumber ν_{crit} and the associated critical activity parameter $\kappa_{\text{crit}} = \kappa(\nu_{\text{crit}})$, we minimize this expression with respect to ν , or equivalently with respect to $x = \nu^2$, which requires that

$$\frac{\partial \lambda}{\partial x}(\nu_{\text{crit}}) = 0. \quad (\text{C9})$$

1. Case A: $q_0 = 0$ and L small

When $c_0 > 0$, the spontaneous order in the uniform steady-state vanishes and we have

$$\lambda = \frac{(x^{-1} + 4)(x + k_d)(2c_0 + 2Lx)}{2c_0 + 2Lx + \kappa\beta(x + k_d)}. \quad (\text{C10})$$

Given a set of parameters, this equation can be numerically minimized to obtain the critical wavenumber and activity. To obtain workable explicit expressions, we first assume that $L = 0$, leading to

$$\lambda = \frac{(1 + 4x)(k_d + x)}{\delta x^2 + (1 + k_d\delta)x}, \quad (\text{C11})$$

where

$$\delta = \frac{\kappa\beta}{2c_0}. \quad (\text{C12})$$

From the condition $0 = \partial\lambda/\partial x$ we obtain

$$\nu_{\text{crit}}^2 = \frac{k_d\delta \pm \sqrt{k_d[4 + \delta(4k_d - 1)]}}{4 - \delta}. \quad (\text{C13})$$

In the limit of $\delta \rightarrow 0$ we recover for the largest root the results $\nu_{\text{crit}}^2 = \nu_{\text{crit},0}^2 = \sqrt{k_d}/2$ and

$\lambda_{\text{crit}} = \lambda_{\text{crit},0} = (1 + 2\sqrt{k_d})^2$ of an active gel model without nematic order [29].

Taylor-expanding the equation above in terms of δ , we obtain

$$\nu_{\text{crit}}^2 = \nu_{\text{crit},0}^2 \left[1 + \frac{1}{8} (1 + 2\sqrt{k_d})^2 \delta \right] + \mathcal{O}(\delta^2), \quad (\text{C14})$$

and

$$\lambda_{\text{crit}} = \lambda_{\text{crit},0} \left[1 - \frac{\sqrt{k_d}}{2} (1 + 2\sqrt{k_d}) \delta \right] + \mathcal{O}(\delta^2). \quad (\text{C15})$$

We note that these expressions are non-dimensional. The dimensional expressions are reported in the main text.

2. Case B: $q_0 = 0$ and c_0 small

Using Eq. (C10), we examine the situation in which c_0 approaches 0 but is positive so that $q_0 = 0$. Eq. (C10) then becomes

$$\lambda = \frac{(1 + 4x)(k_d + x)}{(1 + \delta)x + k_d\delta}, \quad (\text{C16})$$

where now

$$\delta = \frac{\kappa\beta}{2L}. \quad (\text{C17})$$

Following the same rationale as before, we find

$$\nu_{\text{crit}}^2 = \frac{-2k_d\delta + \sqrt{k_d[1 + \delta(1 - 4k_d)]}}{2(1 + \delta)}. \quad (\text{C18})$$

Expanding up to linear order in δ , we obtain

$$\nu_{\text{crit}}^2 = \nu_{\text{crit},0}^2 \left[1 - \frac{1}{2} (1 + 2\sqrt{k_d})^2 \delta \right] + \mathcal{O}(\delta^2), \quad (\text{C19})$$

and

$$\lambda_{\text{crit}} = \lambda_{\text{crit},0} \left[1 - (1 + 2\sqrt{k_d}) \delta \right] + \mathcal{O}(\delta^2). \quad (\text{C20})$$

3. Case C: $q_0 \neq 0$ and L small

Going back to Eq. (C8) and taking $L = 0$ to obtain workable expressions, we obtain

$$\lambda = \frac{(1 + 4x)(x + k_d)}{\delta x^2 + (1 + \tau + \delta k_d)x}, \quad (\text{C21})$$

where

$$\delta = \frac{\kappa\beta}{16bq_0^2} \quad \text{and} \quad \tau = \kappa \left(\frac{\lambda_\odot}{8bq_0} + q_0 \right). \quad (\text{C22})$$

Minimization of κ with respect to x leads to

$$\nu_{\text{crit}}^2 = \frac{\delta k_d + \sqrt{\delta^2 k_d^2 + k_d(4 + 4\tau - \delta)(1 + \tau + \delta k_d)}}{4 + 4\tau - \delta}, \quad (\text{C23})$$

valid when $1 + \tau > 0$. Taylor-expanding the equation above in terms of δ we obtain

$$\nu_{\text{crit}}^2 = \nu_{\text{crit},0}^2 \left[1 + \frac{1}{8} \frac{(1 + 2\sqrt{k_d})^2}{1 + \tau} \delta \right] + \mathcal{O}(\delta^2), \quad (\text{C24})$$

and

$$\lambda_{\text{crit}} = \frac{\lambda_{\text{crit},0}}{1 + \tau} \left[1 - \frac{\sqrt{k_d}}{2} \frac{1 + 2\sqrt{k_d}}{1 + \tau} \delta \right] + \mathcal{O}(\delta^2). \quad (\text{C25})$$

Appendix D: Choice of active gel model parameters in the main text

We consider reference model parameters for the actin cytoskeleton and variations about these values since they can be expected to change significantly between cells and conditions. We take a cortical viscosity of $\eta = 10^4$ Pa s, and hydrodynamic lengths in the order of $l_s = 0.2$ μm [29, 33], although we vary this parameter, as indicated in **Tables I** and **II**. Taking $k_d \sim 0.1 \text{ s}^{-1}$ [26, 29, 96] and $D \sim 0.04 \text{ } \mu\text{m}^2 \text{ s}^{-1}$ [26, 29], we obtain Damköhler lengths l_D in the order of a micron. We consider $\eta_{\text{rot}}/\eta \sim 1$ and $\beta/\eta \sim -0.2$, far from the threshold given by the entropy production inequality. We view κ as the master activity parameter and take values a bit higher than the threshold for pattern formation, which depends on other material parameters as shown in our expressions for κ_{crit} . We vary the nondimensional active tension anisotropy parameter κ . We choose the susceptibility parameters as $a/\eta_{\text{rot}} = 5 \text{ s}^{-1}$ and $b/\eta_{\text{rot}} = 20 \text{ s}^{-1}$ so that relaxation of nematic order is significantly faster than the turnover time-scale and choose the Frank constant to be small so that l_q is small compared to other lengthscales but not too small so that our computational grid can resolve it. The full list of material parameters for each figure and movie is given in the **Tables I** and **II**.

Appendix E: Discrete network simulations

To ascertain the microstructural origin of the anisotropic active tensions underlying the formation of stress fibers, and provide evidence for the orientational activity parameter ($\rho\kappa_0$) included in our theory, we establish an agent-based microscopic model of a crosslinked actomyosin network using the open source cytoskeletal simulation suite cytosim [89, 97]. We customized the source code to impose and track nematic order in the system [98]. Below we provide a concise description of the modeling approach and an overview of the simulation parameters.

1. Model description

Actin filaments are represented through a Langevin equation that recreates the Brownian motion of a thermal system with temperature T (thermal energy: kT , where $k = 1.38 \times 10^{-5}$ pN $\mu\text{m K}^{-1}$ is Boltzmann's constant) and includes bending elasticity as well as external forces (e.g. those applied by crosslinkers and myosin motors). The filaments are immersed in a medium of viscosity ν and the motion of their points is described by

$$d\mathbf{X} = \mu \mathbf{F}(\mathbf{X}, t) dt + d\mathbf{B}(t), \quad (\text{E1})$$

where $\mathbf{F}(\mathbf{X}, t)$ contains the forces acting on the points at time t , $d\mathbf{B}(t)$ is a stochastic non-differentiable function of time summarizing the random molecular collisions that lead to Brownian motions, and the matrix μ contains the mobility coefficients of the particles that constitute the system.

The simulated system comprises N_A actin filaments, which are modeled as inextensible objects with bending rigidity κ_A and individual filament length l_A ; both quantities are uniform across the system. Following a previously-proposed approach [99], we represent actin turnover by randomly selecting and completely removing N_t filaments every N_s time steps and replacing them with an equal number of new ones that are placed at randomly-chosen locations and have the same orientations as the removed filaments. The values of N_t and N_s are chosen to ensure that a given fraction of the total filaments in the system be replaced per unit of time, according to a turnover rate $r_A = N_t/(N_s N_A)$.

To control the nematic characteristics of the filament network, we introduce a system-wide restraining energy of stiffness κ_S that penalizes deviations from a target orientational order parameter S_0 , measured with respect to a director, $\hat{n}$, that is aligned with the Cartesian basis vector $\hat{e}_2$, see **Supp. Fig. 3I(a)**,

$$E_S(\mathbf{X}) = \frac{\kappa_S}{2} \left| \mathbf{q}^s(\mathbf{X}) - \frac{S_0}{2} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \right|^2. \quad (\text{E2})$$

In this equation, we evaluate the average nematic order of the simulation box as a sample average given as $q_{ij}^s = \sum_{I=1}^N (m_i^I m_j^I - \delta_{ij}/2) / N$ over the ensemble of N segments that constitute the filament network and m^I denotes the unit vector corresponding to the I -th segment in the system. The energy defined by **Eq. (E2)** results in restraining forces that act on the particles comprising the actin filaments, such that a particle of position $\mathbf{X}^P$ is subjected to a force

$$\mathbf{f}^P = \mathbf{f}(\mathbf{X}^P) = - \frac{\partial E_S(\mathbf{X})}{\partial \mathbf{X}^P} = - \frac{\partial E_S(\mathbf{X})}{\partial \mathbf{q}^s(\mathbf{X})} \frac{\partial \mathbf{q}^s(\mathbf{X})}{\partial \mathbf{X}^P} = - \kappa_S \left[\mathbf{q}^s(\mathbf{X}) - \frac{S_0}{2} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \right] \frac{\partial \mathbf{q}^s(\mathbf{X})}{\partial \mathbf{X}^P}. \quad (\text{E3})$$

In order to write the derivative $\partial \mathbf{q}^s(\mathbf{X})/\partial \mathbf{X}^P$ explicitly, we denote with $\mathbf{m}^{P,-} = (\mathbf{X}^P - \mathbf{X}^{P-})/|\mathbf{X}^P - \mathbf{X}^{P-}|$ and $\mathbf{m}^{P,+} = (\mathbf{X}^{P+1} - \mathbf{X}^P)/|\mathbf{X}^{P+1} - \mathbf{X}^P|$ the unit vectors that correspond to the filament segments extending from point $\mathbf{X}^P$ towards the head (+) and the tail (-) of the filament, see **Supp. Fig. 3I(b)**. In the absence of filament bifurcations, which are not present in our model, $\mathbf{m}^{P,+}$ and $\mathbf{m}^{P,-}$ are the only unit vectors in the network that depend on $\mathbf{X}^P$. Thus, it is immediate to simplify the derivative on the right hand side of **Eq. (E3)** and express the restraining force acting on the P -th particle as

$$\mathbf{f}^P = -\frac{\kappa_S}{N} \left[q_{ij}^s - \frac{S_0}{2} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \right] \begin{pmatrix} \frac{\partial m_i^{P,-}}{\partial X_j^P} m_j^{P,-} + m_i^{P,-} \frac{\partial m_j^{P,-}}{\partial X_j^P} + \frac{\partial m_i^{P,+}}{\partial X_j^P} m_j^{P,+} + m_i^{P,+} \frac{\partial m_j^{P,+}}{\partial X_j^P} \\ \frac{\partial m_i^{P,-}}{\partial X_j^P} m_j^{P,-} + m_i^{P,-} \frac{\partial m_j^{P,-}}{\partial X_j^P} + \frac{\partial m_i^{P,+}}{\partial X_j^P} m_j^{P,+} + m_i^{P,+} \frac{\partial m_j^{P,+}}{\partial X_j^P} \end{pmatrix}. \quad (\text{E4})$$

We shall also note that **Eq. (E4)** is simplified when particle P coincides with the head or the tail of a filament, since either $\mathbf{m}^{P,+}$ or $\mathbf{m}^{P,-}$ will not exist. For a filament head, the restraining energy introduced in **Eq. (E2)** will thus result in the force

$$\mathbf{f}_{\text{head}}^P = -\frac{\kappa_S}{N} \left[q_{ij}^s - \frac{S_0}{2} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \right] \begin{pmatrix} \frac{\partial m_i^{P,-}}{\partial X_j^P} m_j^{P,-} + m_i^{P,-} \frac{\partial m_j^{P,-}}{\partial X_j^P} \\ \frac{\partial m_i^{P,-}}{\partial X_j^P} m_j^{P,-} + m_i^{P,-} \frac{\partial m_j^{P,-}}{\partial X_j^P} \end{pmatrix}. \quad (\text{E5})$$

Similarly, the force acting at a filament tail will be

$$\mathbf{f}_{\text{tail}}^P = -\frac{\kappa_S}{N} \left[q_{ij}^s - \frac{S_0}{2} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \right] \begin{pmatrix} \frac{\partial m_i^{P,+}}{\partial X_j^P} m_j^{P,+} + m_i^{P,+} \frac{\partial m_j^{P,+}}{\partial X_j^P} \\ \frac{\partial m_i^{P,+}}{\partial X_j^P} m_j^{P,+} + m_i^{P,+} \frac{\partial m_j^{P,+}}{\partial X_j^P} \end{pmatrix}. \quad (\text{E6})$$

Finally, assuming that each actin filament is represented by segments of constant length s_A , the derivatives of the unit vectors appearing in Eqs. (E4) – (E6) are obtained directly from the definitions of $\mathbf{m}^{P,-}$, and $\mathbf{m}^{P,+}$ as

$$\begin{aligned} \frac{\partial m_i^{P,-}}{\partial X_j^P} &= \frac{(X_j^P - X_j^{P-1})^2}{s_A^3}, & \frac{\partial m_j^{P,-}}{\partial X_i^P} &= -\frac{(X_i^P - X_i^{P-1})(X_j^P - X_j^{P-1})}{s_A^3}, \\ \frac{\partial m_i^{P,-}}{\partial X_j^P} &= -\frac{(X_i^P - X_i^{P-1})(X_j^P - X_j^{P-1})}{s_A^3}, & \frac{\partial m_j^{P,-}}{\partial X_j^P} &= \frac{(X_i^P - X_i^{P-1})^2}{s_A^3}, \\ \frac{\partial m_i^{P,+}}{\partial X_j^P} &= -\frac{(X_j^{P+1} - X_j^P)^2}{s_A^3}, & \frac{\partial m_j^{P,+}}{\partial X_i^P} &= \frac{(X_i^{P+1} - X_i^P)(X_j^{P+1} - X_j^P)}{s_A^3}, \\ \frac{\partial m_i^{P,+}}{\partial X_j^P} &= \frac{(X_i^{P+1} - X_i^P)(X_j^{P+1} - X_j^P)}{s_A^3}, & \frac{\partial m_j^{P,+}}{\partial X_j^P} &= -\frac{(X_i^{P+1} - X_i^P)^2}{s_A^3}. \end{aligned} \quad (\text{E7})$$

Crosslinkers and myosin motors are modeled as Hookean springs with finite stiffness K_X and K_M and resting lengths ℓ_X and ℓ_M , respectively. For simplicity, the concentration of unattached species is assumed to be uniform across the modeling space and their diffusion is thus not simulated explicitly. The ends of the springs can bind to any discrete location along the filament segments, as long as they fall within finite binding ranges denoted as d_X^b and d_M^b . When multiple filament locations fall within the binding range, the springs attach to the closest point on the filament and apply a force that depends linearly on their elongation (u_X and u_M)

$$\begin{aligned} f_X &= K_X (u_X - \ell_X), \\ f_M &= K_M (u_M - \ell_M). \end{aligned} \quad (\text{E8})$$

Binding of the species is modeled as a purely stochastic event whose time of occurrence follows an exponential distribution. The expected values for crosslinker and motor binding are $1/r_X^b$ and $1/r_M^b$, r_X^b and r_M^b being the corresponding binding rates. Unbinding from a filament is not always purely stochastic but follows a rate that can increase when the springs are loaded, according to a relationship known as Bell's law [100]. Denoting the base unbinding rates for crosslinkers and motors as $r_X^{u,0}$ and $r_M^{u,0}$, the effective rates under a force of magnitude f are expressed as

$$\begin{aligned} r_X^u &= r_X^{u,0} e^{f/f_X^u}, \\ r_M^u &= r_M^{u,0} e^{f/f_M^u}, \end{aligned} \quad (\text{E9})$$

where f_X^u and f_M^u are constant parameters associated with the bound state. A specific feature of motors is that they can 'walk' on actin filaments by displacing their attachment points without necessarily having to unbind; this effect can be modified by load application. Denoting with $v_M^{\max}$ the speed of the motor in the absence of external loads, the effective speed is given by

$$v_M = v_M^{\max} (1 + f/f_M^{\text{stall}}), \quad (\text{E10})$$

where f_M^{stall} is the stall force, which controls the slope of the speed-force relationship. By convention, we consider that positive (negative) speed values correspond to a motion towards the head (tail) of a filament.

2. Quantification of network-scale active tensions

To determine anisotropic active tensions arising when the network is driven out-of-equilibrium, we examine a square representative volume element of crosslinked actomyosin network at room temperature ($kT = 0.0042$ pN μm). The modeled systems feature an edge length 5 μm , a filament surface density ρ_A , a crosslinker density $\rho_X = 15.4$ (actin μm)⁻¹, and a myosin motor density $\rho_M = 0.8$ (actin μm)⁻¹. The actin density ρ_A is varied throughout the study and takes the following values: $\rho_A = (78; 156; 234)$ (actin μm)/ μm^2 , which we denote as ρ , 2ρ , and 3ρ . All simulation parameters adopted in this study are indicated in **Tables III** – **V**, where we also provide an overview of the values used in previously-proposed microstructural models that are comparable to the ones developed here. Throughout the simulations, we adopt a time step of 5 ms and a data acquisition frequency of 40 s⁻¹, i.e. 1 every 5 time steps is written to output for further data processing.

The filaments are initially seeded uniformly throughout the modeling space and without any orientational bias, **Supp. Fig. 3II(a)**, followed by equilibration in the presence of the restraining energy defined by **Eq. (E2)**, which acts with a stiffness $\kappa_S = 5000$ pN/ μm to induce a nematic orientation characterized by the ordering parameter S_0 , see **Supp. Fig. 3II(b)**. No crosslinkers or myosin motors are present at this stage. After 5 s, **Supp. Fig. 3II(c)**, the filaments are clamped to the system's boundary using N_H anchors. These objects, whose spatial position is fixed, are akin to crosslinkers but have spring stiffness $K_H = 5000$ pN/ μm , zero resting length, a binding range $d_H^b = 5$ nm, and a binding rate $r_H^b = 100$ s⁻¹. Anchor unbinding is completely hindered by setting $r_H^{u,0} = 0$ s⁻¹ and $f_H^u = \infty$. The quantity N_H is defined by the number of filaments that cross the system's boundaries. Indicating with N_H^L , N_H^R , N_H^B , and N_H^T the number of anchoring objects located on the left, right, bottom, and top edge of the system, it follows that $N_H = N_H^L + N_H^R + N_H^B + N_H^T$.

After having equilibrated the network for an additional 2.5 s, we deactivate the restraining potential by setting $\kappa_S = 0$ and introduce crosslinkers and motors in the system, **Supp. Fig. 3II(d)**, which we let interact with the filaments for 12.5 s in order to drive the system out-of-

equilibrium. This results in reaction forces being applied at the anchoring points. At each time step, the total average tensions acting on the system edges that are oriented parallel and perpendicular to the nematic director $\hat{n}$, $\sigma_{||}$ and $\sigma_{\perp}$, are measured as

$$\begin{aligned}\sigma_{||} &= \frac{1}{2L} \left(\sum_{I=1}^{N_H^T} \mathbf{F}_I^T \cdot \hat{\mathbf{e}}_2 - \sum_{I=1}^{N_H^B} \mathbf{F}_I^B \cdot \hat{\mathbf{e}}_2 \right), \\ \sigma_{\perp} &= \frac{1}{2L} \left(\sum_{I=1}^{N_H^R} \mathbf{F}_I^R \cdot \hat{\mathbf{e}}_1 - \sum_{I=1}^{N_H^L} \mathbf{F}_I^L \cdot \hat{\mathbf{e}}_1 \right).\end{aligned}\tag{E11}$$

Measuring tensions from, $t = 10$ s, **Supp. Fig. 3II(e)**, to $t = 20$ s, **Supp. Fig. 3II(f)**, allows us to quantify tension anisotropy using the deviatoric tension normalized by the mean tension

$$\xi = 2 \frac{\langle \sigma_{||} - \sigma_{\perp} \rangle}{\langle \sigma_{||} + \sigma_{\perp} \rangle},\tag{E12}$$

where $\langle \cdot \rangle$ denotes the time average over the considered period of interest. Finally, quantifying the above ratio for 12 sets of $n=16$ model realizations that correspond to values of $S_0 = (0.0; 0.1; 0.2; 0.3)$ and $\rho_A = (\rho; 2\rho; 3\rho)$ allows us to determine the mesoscale activity parameter κ by fitting the ξ versus S_0 relation, see **Fig. 5e**. To this end, we leverage the function ‘stats.linregress’ that is included in the Python package ‘SciPy’ [101].

3. Quantification of the orientational activity parameter

To provide evidence for the orientational activity parameter in our theory ($\rho\lambda_{\odot}$), we examine the behavior of a periodic square representative volume element of crosslinked actomyosin driven out-of-equilibrium. The absence of anchors in these systems allows us to track the unconstrained nematic order dynamics. We limit our analysis to models with actin density $\rho_A = 2\rho$ and adopt the same simulation parameters as in Section E 2, unless explicitly stated otherwise.

To highlight the contribution of the restraining potential in Eq. (E2), we initially focus on athermal systems ($\kappa T \approx 0$). After seeding N_A actin filaments uniformly throughout the modeling space and without any orientational bias, **Supp. Fig. 3III(a)**, we let the restraining potential reorient the system for 2.5 s in the absence of crosslinkers and myosin motors to reach an ordering parameter, S_0 **Supp. Fig. 3III(b)**. Following the first 2.5 s of athermal simulation, we increase the temperature to its standard value ($kT = 0.0042$ pN μm), deactivate the restraining potential by setting $\kappa_S = 0$, and add crosslinkers and myosin motors to the system, which we let interact with the filaments for 30 s in order to drive the system out-of-equilibrium, **Supp. Fig. 3III(c,d)**. We observe that $S(t)$ increases monotonically, allowing us to estimate $\dot{S}$ according to a linear least-squares fit, as implemented in the Python ‘SciPy’ function ‘stats.linregress’ [101]. We notice that in **Supp. Fig. 3III(a,b)** the time evolution of S is approximately exponential, **Supp. Fig. 3III(e)**, as predicted in the main text. This allows us to leverage our discrete network simulations to estimate the model parameter $\eta_{\text{rot}} \approx 1583$ pN μm s using the nonlinear least-squares approach provided by the function ‘optimize.curvefit’ that is included in the Python package ‘SciPy’ [101]. Note that the value of η_{rot} is determined by fitting one single evolution curve for $S(t)/S_0$, which is obtained by averaging the system dynamics resulting from the 4 sets of $n = 16$ model realizations that correspond to values of $S_0 = (0.1; 0.2; 0.3; 0.4)$.

Finally, we plot our simulation-based approximation of $\dot{S}$ for several values of S_0 , which results in an almost linear relationship with positive slope, implying that $\rho\lambda_{\odot} > 2a$ (see **Fig. 5g**). Moreover, considering fully athermal simulations (i.e. simulations in which the temperature is not increased after reaching the ordering parameter S_0) results in a linear relation between $\dot{S}$ and S_0 with

slightly larger slope. This finding agrees with the predictions of our theory for $a > 0$ and demonstrates that temperature, which entropically drives the network towards isotropy, opposes the effect of the orientational activity parameter $\rho\lambda_{\odot}$.

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Reviewer #1 (Public Review):

Summary: In this article, Mirza et al developed a continuum active gel model of actomyosin cytoskeleton that account for nematic order and density variations in actomyosin. Using this model, they identify the requirements for the formation of dense nematic structures. In

particular, they show that self-organization into nematic bundles requires both flow-induced alignment and active tension anisotropy in the system. By varying model parameters that control active tension and nematic alignment, the authors show that their model reproduces a rich variety of actomyosin structures, including tactoids, fibres, asters as well as crystalline networks. Additionally, discrete simulations are employed to calculate the activity parameters in the continuum model, providing a microscopic perspective on the conditions driving the formation of fibrillar patterns.

Strengths: The strength of the work lies in its delineation of the parameter ranges that generate distinct types of nematic organization within actomyosin networks. The authors pinpoint the physical mechanisms behind the formation of fibrillar patterns, which may offer valuable insights into stress fiber assembly. Another strength of the work is connecting activity parameters in the continuum theory with microscopic simulations.

Weaknesses: This paper is a very difficult read for nonspecialists, especially if you are not well-versed in continuum hydrodynamic theories. Efforts should be made to connect various elements of theory with biological mechanisms, which is mostly lacking in this paper. The comparison with experiments is predominantly qualitative. It is unclear if the theory is suited for in vitro or in vivo actomyosin systems. The justification for various model assumptions, especially concerning their applicability to actomyosin networks, requires a more thorough examination. The classification of different structures demands further justification. For example, the rationale behind categorizing structures as sarcomeric remains unclear when nematic order is perpendicular to the axis of the bands. Sarcomeres traditionally exhibit a specific ordering of actin filaments with alternating polarity patterns. Similarly, the criteria for distinguishing between contractile and extensile structures need clarification, as one would expect extensile structures to be under tension contrary to the authors' claim. Additionally, it is unclear if the model's predictions for fiber dynamics align with observations in cells, as stress fibers exhibit a high degree of dynamism and tend to coalesce with neighboring fibers during their assembly phase. Finally, it seems that the microscopic model is unable to recapitulate the density patterns predicted by the continuum theory, raising questions about the suitability of the simulation model.

<https://doi.org/10.7554/eLife.93097.2.sa2>

Reviewer #2 (Public Review):

Summary:

The article by Waleed et al discusses the self organization of actin cytoskeleton using the theory of active nematics. Linear stability analysis of the governing equations and computer simulations show that the system is unstable to density fluctuations and self organized structures can emerge. While the context is interesting, I am not sure whether the physics is new. Hence I have reservations about recommending this article.

Strengths:

(i) Analytical calculations complemented with simulations (ii) Theory for cytoskeletal network

Weaknesses:

Not placed in the context or literature on active nematics.

<https://doi.org/10.7554/eLife.93097.2.sa1>

Reviewer #3 (Public Review):

The manuscript "Theory of active self-organization of dense nematic structures in the actin cytoskeleton" analysis self-organized pattern formation within a two-dimensional nematic liquid crystal theory and uses microscopic simulations to test the plausibility of some of the conclusions drawn from that analysis. After performing an analytic linear stability analysis that indicates the possibility of patterning instabilities, the authors perform fully non-linear numerical simulations and identify the emergence of stripe-like patterning when anisotropic active stresses are present. Following a range of qualitative numerical observations on how parameter changes affect these patterns, the authors identify, besides isotropic and nematic stress, also active self-alignment as an important ingredient to form the observed patterns. Finally, microscopic simulations are used to test the plausibility of some of the conclusions drawn from continuum simulations.

The paper is well written, figures are mostly clear and the theoretical analysis presented in both, main text and supplement, is rigorous. Mechano-chemical coupling has emerged in recent years as a crucial element of cell cortex and tissue organization and it is plausible to think that both, isotropic and anisotropic active stresses, are present within such effectively compressible structures. Even though not yet stated this way by the authors, I would argue that combining these two is of the key ingredients that distinguishes this theoretical paper from similar ones. The diversity of patterning processes experimentally observed is nicely elaborated on in the introduction of the paper, though other closely related previous work could also have been included in these references (see below for examples).

To introduce the continuum model, the authors exclusively cite their own, unpublished pre-print, even though the final equations take the same form as previously derived and used by other groups working in the field of active hydrodynamics (a certainly incomplete list: Marenduzzo et al (PRL, 2007), Salbreux et al (PRL, 2009, cited elsewhere in the paper), Jülicher et al (Rep Prog Phys, 2018), Giomi (PRX, 2015),...). To make better contact with the broad active liquid crystal community and to delineate the present work more compellingly from existing results, it would be helpful to include a more comprehensive discussion of the background of the existing theoretical understanding on active nematics. In fact, I found it often agrees nicely with the observations made in the present work, an opportunity to consolidate the results that is sometimes currently missed out on. For example, it is known that self-organised active isotropic fluids form in 2D hexagonal and pulsatory patterns (Kumar et al, PRL, 2014), as well as contractile patches (Mietke et al, PRL 2019), just as shown and discussed in Fig. 2. It is also known that extensile nematics, $\kappa < 0$ here, draw in material laterally of the nematic axis and expel it along the nematic axis (the other way around for $\kappa > 0$, see e.g. Doostmohammadi et al, Nat Comm, 2018 "Active Nematics" for a review that makes this point), consistent with all relative nematic director/flow orientations shown in Figs. 2 and 3 of the present work.

The results of numerical simulations are well-presented. Large parts of the discussion of numerical observations - specifically around Fig. 3 - are qualitative and it is not clear why the analysis is restricted to $\kappa < 0$. Some of the observations resonate with recent discussions in the field, for example the observation of effectively extensile dynamics in a contractile system is interesting and reminiscent of ambiguities about extensile/contractile properties discussed in recent preprints (<https://arxiv.org/abs/2309.04224>). It is convincingly concluded that, besides nematic stress on top of isotropic one, active self-alignment is a key ingredient to produce the observed patterns.

I compliment the authors for trying to gain further mechanistic insights into this conclusion with microscopic filament simulations that are diligently performed. It is rightfully stated that these simulations only provide plausibility tests and, within this scope, I would say the authors are successful. At the same time, it leaves open questions that could have been discussed more carefully. For example, I wonder what can be said about the regime $\kappa > 0$

(which is dropped ad-hoc from Fig. 3 onward) microscopically, in which the continuum theory does also predict the formation of stripe patterns - besides the short comment at the very end? How does the spatial inhomogeneous organization the continuum theory predicts fit in the presented, microscopic picture and vice versa?

Overall, the paper represents a valuable contribution to the field of active matter and, if strengthened further, might provide a fruitful basis to develop new hypothesis about the dynamic self-organisation of dense filamentous bundles in biological systems.

<https://doi.org/10.7554/eLife.93097.2.sa0>

Author Response

eLife assessment

In this study, the authors offer a theoretical explanation for the emergence of nematic bundles in the actin cortex, carrying implications for the assembly of actomyosin stress fibers. As such, the study is a valuable contribution to the field actomyosin organization in the actin cortex. While the theoretical work is solid, experimental evidence in support of the model assumptions remains incomplete. The presentation could be improved to enhance accessibility for readers without a strong background in hydrodynamic and nematic theories.

To address the weaknesses identified in this assessment, we plan to expand the description of the theoretical model to make it more accessible to a broader spectrum of readers. We will discuss in more detail the relation between the different mathematical terms and physical processes at the molecular scale, as well as the experimental evidence supporting the model assumptions. We will also discuss more explicitly how our results are relevant to different systems exhibiting actomyosin nematic bundles beyond stress fibers.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

In this article, Mirza et al developed a continuum active gel model of actomyosin cytoskeleton that account for nematic order and density variations in actomyosin. Using this model, they identify the requirements for the formation of dense nematic structures. In particular, they show that self-organization into nematic bundles requires both flow-induced alignment and active tension anisotropy in the system. By varying model parameters that control active tension and nematic alignment, the authors show that their model reproduces a rich variety of actomyosin structures, including tactoids, fibres, asters as well as crystalline networks. Additionally, discrete simulations are employed to calculate the activity parameters in the continuum model, providing a microscopic perspective on the conditions driving the formation of fibrillar patterns.

Strengths:

The strength of the work lies in its delineation of the parameter ranges that generate distinct types of nematic organization within actomyosin networks. The authors pinpoint the physical mechanisms behind the formation of fibrillar patterns, which may offer valuable insights into stress fiber assembly. Another strength of the work is connecting activity parameters in the continuum theory with microscopic simulations.

We thank the referee for these comments.

Weaknesses:

This paper is a very difficult read for nonspecialists, especially if you are not well-versed in continuum hydrodynamic theories. Efforts should be made to connect various elements of theory with biological mechanisms, which is mostly lacking in this paper. The comparison with experiments is predominantly qualitative.

We agree with the referee that the manuscript will benefit from a better description of the theoretical model and the results in relation with specific molecular and cellular mechanisms. We will further emphasize how a number of experimental observations in the literature support our model assumptions and can be explained by our results. A quantitative comparison is difficult for several reasons. First, many of the parameters in our theory have not been measured, and in fact estimates in the literature often rely on comparison with hydrodynamic models such as ours. Second, the effective physical properties of actomyosin gels can vary wildly between cells, which may explain the diversity of forms, dynamics and functions. For these reasons, we chose to delineate regimes leading to qualitatively different emerging architectures and dynamics. In the revised manuscript, we will make this point clearer and will further study the literature to seek quantitative comparison.

It is unclear if the theory is suited for in vitro or in vivo actomyosin systems. The justification for various model assumptions, especially concerning their applicability to actomyosin networks, requires a more thorough examination.

We thank the referee for this comment. Our theory is applicable to actomyosin in living cells. To our knowledge, reconstituted actomyosin gels currently lack the ability to sustain the dynamical steady-states involved in the proposed self-organization mechanism, which balance actin flows with turnover. In addition to actomyosin gels in living cells, in vitro systems based on encapsulated cell extracts can also sustain such dynamical steady states [e.g. <https://doi.org/10.1038/s41567-018-0413-4>], and therefore our theory may be applicable to these systems as well. Of course, with advancements in the field of reconstituted systems, this may change in the near future. We will explicitly discuss this point in the revised manuscript.

The classification of different structures demands further justification. For example, the rationale behind categorizing structures as sarcomeric remains unclear when nematic order is perpendicular to the axis of the bands. Sarcomeres traditionally exhibit a specific ordering of actin filaments with alternating polarity patterns.

We agree and will avoid the term “sarcomeric”.

Similarly, the criteria for distinguishing between contractile and extensile structures need clarification, as one would expect extensile structures to be under tension contrary to the authors' claim.

We plan to clarify this point by representing in a main figure the stress profiles across dense nematic structures (currently in Supp Fig 2), along with a more detailed description. In short, depending on the parameter regime, the competition between active and viscous stresses in the actin gel determine whether the emergent structures are extensile or contractile. In our system tension is positive in all directions at all times. However, in “contractile” structures, tension is larger along the bundle, whereas in “extensile” structures, tension is larger perpendicular to the bundle. This is consistent with the common expression for active stress of incompressible nematic systems [see e.g. <https://doi.org/10.1038/s41467-018-05666-8>], that takes the form $-zQ$, where z is positive for an extensile system, showing that in this case

active tension is negative along the nematic direction. This point, also been raised by another referee, will be clarified and connected to existing literature.

Additionally, its unclear if the model's predictions for fiber dynamics align with observations in cells, as stress fibers exhibit a high degree of dynamism and tend to coalesce with neighboring fibers during their assembly phase.

In the present work, we focus on the self-organization of a periodic patch of actomyosin gel. However, in adherent cells boundary conditions play an essential role, e.g. with inflow at the cell edge as a result of polymerization and exclusion at the nucleus. In ongoing work, we are studying with the present model the dynamics of assembly and reconfiguration of dense nematic structures in domains with boundary conditions mimicking in adherent cells, as suggested by the referee. We would like to note, however, that the prominent stress fibers in cells adhered to stiff substrates, so abundantly reported in the literature, are not the only instance of dense nematic actin bundles, and may not be representative of physiologically relevant situations. In the present manuscript, we emphasize the relation of the predicted organizations with those found in different in vivo contexts not related to stress fibers, such as the aligned patterns of bundles in insects (trachea, scales in butterfly wings), in hydra, or in reproductive organs of *C. elegans*; the highly dynamical network of bundles observed in *C. elegans* early embryos; or the labyrinth patterns of micro-ridges in the apical surface of epidermal cells in fish. We will further emphasize these points in the revised manuscript.

Finally, it seems that the microscopic model is unable to recapitulate the density patterns predicted by the continuum theory, raising questions about the suitability of the simulation model.

We thank the referee for raising this question, which needs further clarification. The goal of the microscopic model is not to reproduce the self-organized patterns predicted by the active gel theory. The microscopic model lacks essential ingredients, notably a realistic description of hydrodynamics and turnover. Our goal with the agent-based simulations is to extract the relation between nematic order and active stresses for a small homogeneous sample of the network. This small domain is meant to represent the homogeneous active gel prior to pattern formation, and it allows us to substantiate key assumptions of the continuum model leading to pattern formation, notably the dependence of isotropic and deviatoric components of the active stress on density and nematic order (Eq. 7) and the active generalized stress promoting ordering.

We should mention that reproducing the range of out-of-equilibrium mesoscale architectures predicted by our active gel model with agent-based simulations seems at present not possible, or at least significantly beyond the state-of-the-art. We note for instance that parameter regimes in which agent-based simulations of actin gels display extended contractile steady-states are non-generic, as these simulations often lead to irreversible clumping (as do many reconstituted contractile systems), see e.g. <https://doi.org/10.1038/ncomms10323> or <https://doi.org/10.1371/journal.pcbi.1005277>. Very few references report sustained actin flows or the organization of a few bundles (<https://doi.org/10.1371/journal.pcbi.1009506>). While agent-based cytoskeletal simulations are very attractive because they directly connect with molecular mechanisms, active gel continuum models are better suited to describe out-of-equilibrium emergent hydrodynamics at a mesoscale. We believe that these two complementary modeling frameworks are rather disconnected in the literature, and for this reason, we have attempted substantiate our continuum modeling with discrete simulations. In the revised manuscript, we will better frame the relationship between them.

Reviewer #2 (Public Review):

Summary:

The article by Waleed et al discusses the self organization of actin cytoskeleton using the theory of active nematics. Linear stability analysis of the governing equations and computer simulations show that the system is unstable to density fluctuations and self organized structures can emerge. While the context is interesting, I am not sure whether the physics is new. Hence I have reservations about recommending this article.

We thank the referee for these comments. In the revised manuscript, we will highlight the novelty of the paper in terms of the theoretical model, the mechanism of patterning of dense nematic structures, the nature and dynamics of the resulting architectures, their relation with the experimental record, and the connection with microscopic models.

We will emphasize the fact that nematic architectures in the actin cytoskeleton are characterized by a co-localization of order and density (and strong variations in each of these fields), that recent work shows that isotropic and nematic organizations coexist and are part of a single heterogeneous network, that the emergence and maintenance of nematic order requires active contraction, and that the assembly and maintenance of dense nematic bundles involves convergent flows. None of these key features can be described by the common incompressible models of active nematics. To address this, we develop here a compressible and density dependent model for an active nematic gel. We will carefully justify that the proposed model is meaningful for actomyosin gels, and we will highlight the commonalities and differences with previous models of active nematics.

Strengths:

(i) Analytical calculations complemented with simulations (ii) Theory for cytoskeletal network

Weaknesses:

Not placed in the context or literature on active nematics.

We agree with the referee that the manuscript requires a better contextualization of the work in relation with the very active field of active nematics. In the revised manuscript, we will clearly describe the relation of our model with existing ones.

Reviewer #3 (Public Review):

The manuscript "Theory of active self-organization of dense nematic structures in the actin cytoskeleton" analysis self-organized pattern formation within a two-dimensional nematic liquid crystal theory and uses microscopic simulations to test the plausibility of some of the conclusions drawn from that analysis. After performing an analytic linear stability analysis that indicates the possibility of patterning instabilities, the authors perform fully non-linear numerical simulations and identify the emergence of stripelike patterning when anisotropic active stresses are present. Following a range of qualitative numerical observations on how parameter changes affect these patterns, the authors identify, besides isotropic and nematic stress, also active self-alignment as an important ingredient to form the observed patterns. Finally, microscopic simulations are used to test the plausibility of some of the conclusions drawn from continuum simulations.

The paper is well written, figures are mostly clear and the theoretical analysis presented in both, main text and supplement, is rigorous. Mechano-chemical coupling has emerged in recent years as a crucial element of cell cortex and tissue organization and it is plausible to think that both, isotropic and anisotropic active stresses, are present within such effectively compressible structures. Even though not yet stated this way by the authors, I would argue that combining these two is of the key ingredients that

distinguishes this theoretical paper from similar ones. The diversity of patterning processes experimentally observed is nicely elaborated on in the introduction of the paper, though other closely related previous work could also have been included in these references (see below for examples).

We thank the referee for these comments and for the suggestion to emphasize the interplay of isotropic and anisotropic active tension, which is possible only in a compressible gel. We thank the suggestions of the referee to better connect with existing literature.

To introduce the continuum model, the authors exclusively cite their own, unpublished pre-print, even though the final equations take the same form as previously derived and used by other groups working in the field of active hydrodynamics (a certainly incomplete list: Marenduzzo et al (PRL, 2007), Salbreux et al (PRL, 2009, cited elsewhere in the paper), Jülicher et al (Rep Prog Phys, 2018), Giomi (PRX, 2015),...). To make better contact with the broad active liquid crystal community and to delineate the present work more compellingly from existing results, it would be helpful to include a more comprehensive discussion of the background of the existing theoretical understanding on active nematics. In fact, I found it often agrees nicely with the observations made in the present work, an opportunity to consolidate the results that is sometimes currently missed out on. For example, it is known that self-organised active isotropic fluids form in 2D hexagonal and pulsatory patterns (Kumar et al, PRL, 2014), as well as contractile patches (Mietke et al, PRL 2019), just as shown and discussed in Fig. 2. It is also known that extensile nematics, $\kappa < 0$ here, draw in material laterally of the nematic axis and expel it along the nematic axis (the other way around for $\kappa > 0$, see e.g. Doostmohammadi et al, Nat Comm, 2018 "Active Nematics" for a review that makes this point), consistent with all relative nematic director/flow orientations shown in Figs. 2 and 3 of the present work.

We thank the referee for these suggestions. Indeed, in the original submission we had outsourced much of the justification of the model and the relevant literature to a related pre-print, but this is not reasonable. In the revised manuscript, we will discuss our model in the context of the state-of-the-art, emphasizing connections with existing results.

The results of numerical simulations are well-presented. Large parts of the discussion of numerical observations - specifically around Fig. 3 - are qualitative and it is not clear why the analysis is restricted to $\kappa < 0$. Some of the observations resonate with recent discussions in the field, for example the observation of effectively extensile dynamics in a contractile system is interesting and reminiscent of ambiguities about extensile/contractile properties discussed in recent preprints (<https://arxiv.org/abs/2309.04224>). It is convincingly concluded that, besides nematic stress on top of isotropic one, active self-alignment is a key ingredient to produce the observed patterns.

We thank the referee for these comments. We will expand the description of the results around Figure 3. We are reluctant to extend the detailed analysis of emergent architectures and dynamics to the case $\kappa > 0$ as it leads to architectures not observed, to our knowledge, in actin networks. We will expand the characterization of emergent contractile/extensile networks by describing the distribution of the different components of the stress tensor across the bundles and will place our results in the context of related recent work.

I compliment the authors for trying to gain further mechanistic insights into this conclusion with microscopic filament simulations that are diligently performed. It is rightfully stated that these simulations only provide plausibility tests and, within this scope, I would say the authors are successful. At the same time, it leaves open questions

that could have been discussed more carefully. For example, I wonder what can be said about the regime $\kappa > 0$ (which is dropped ad-hoc from Fig. 3 onward) microscopically, in which the continuum theory does also predict the formation of stripe patterns - besides the short comment at the very end? How does the spatial inhomogeneous organization the continuum theory predicts fit in the presented, microscopic picture and vice versa?

We thank the referee for this compliment. We think that the point raised by the referee is very interesting. It is reasonable to expect that the sign of κ will not be a constant but rather depend on S and ρ . Indeed, for a sparse network with low order, the progressive bundling by crosslinkers acting on nearby filaments is likely to produce a large active stress perpendicular to the nematic direction, whereas in a dense and highly ordered region, myosin motors are more likely to effectively contract along the nematic direction whereas there is little room for additional lateral contraction by additional bundling. In the revised manuscript, we envision to further deepen in this issue in two ways. First, we plan to perform additional agent-based simulations in a regime leading to $\kappa > 0$. Second, we will modify the active gel model such that $\kappa < 0$ for low density/order, so that a fibrillar pattern is assembled, and $\kappa > 0$ for high density/order, so that the emergent fibers are highly contractile.

Overall, the paper represents a valuable contribution to the field of active matter and, if strengthened further, might provide a fruitful basis to develop new hypothesis about the dynamic self-organisation of dense filamentous bundles in biological systems.

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